Economic Transformation

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Summary

Most FGD groups reported that CFM had not improved access to natural resources for either household use or sale, though approximately one third indicated that access had increased. Community youth were significantly less likely than CFMG Executive FGDs to report improved access for household use. Individual interviewees showed a similar pattern, with DFOs significantly more likely than CFMG executives or traditional leaders to consider that CFM had improved household resource access.

Employment outcomes were more positive. Approximately half of FGD groups agreed that CFM had generated new employment opportunities, with a marked contrast between community youth (who were most likely to agree) and community female FGDs (who were most likely to disagree). Across the surveyed CFMGs, reported employment increased substantially after CFM establishment, though there was considerable variation among provinces (means ranging from 3 in Western Province to 21 in Lusaka Province). Employment was strongly male-biased and the dominant employment category was HFO or Community Scout.

Income earning opportunities from CFM remain limited for most communities. Around one third of CFMGs had received carbon finance payments, and awareness of carbon credit agreements was generally high, though youth were significantly less informed than CFMG Executives. Carbon trader support was primarily directed at forest management and agricultural extension. Natural resource value chains — particularly honey and mushrooms — dominated both realised and aspirational income sources. Most forest products are sold only at local markets and private enterprise partnerships were uncommon, with the vast majority involving honey off-takers. Communities identified timber as the most important unrealised commercial opportunity, despite none of the surveyed CFMGs being involved in timber production.

Opportunity costs of CFM were recognised by only about one third of FGD groups. Unexpectedly, non-timber forest product (NTFP) harvesting was frequently cited as a precluded activity, suggesting possible misunderstandings around forest use rules in some communities.

It is evident from the survey that most CFMG business plans are overly reliant on carbon and potentially valuable NTFP value chains remain under-exploited.

Introduction

In this chapter, we consider the effect of CFM management on indicators of economic transformation.

Access to natural resources

We asked FGD groups whether CFM had improved access to natural resources for household use or for sale (Figure 1, Figure 2).

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = case_match(
      as.character(economic_transformation_economic_benefits_from_cfm_improved_access_to_natu
      "1" ~ "Yes",
      "0" ~ "No"
    ),
    response = factor(response, levels = c("Yes", "No")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
  ) |>
  drop_na(response) |>
  count(interview_type, response) |>
  group_by(interview_type) |>
  mutate(prop = n / sum(n)) |>
  ungroup() |>
  ggplot(aes(y = interview_type, x = prop, fill = response)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_text(
    aes(label = ifelse(prop >= 0.05, n, "")),
    position = position_stack(vjust = 0.5),
    size = 3.5, fontface = "bold", colour = "white"
  ) +
```

```r
  scale_x_continuous(
    labels = percent_format(accuracy = 1),
    expand = expansion(mult = 0.01),
    name   = "Proportion of groups"
  ) +
  scale_fill_manual(
    values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
    guide  = guide_legend(reverse = TRUE)
  ) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 14) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

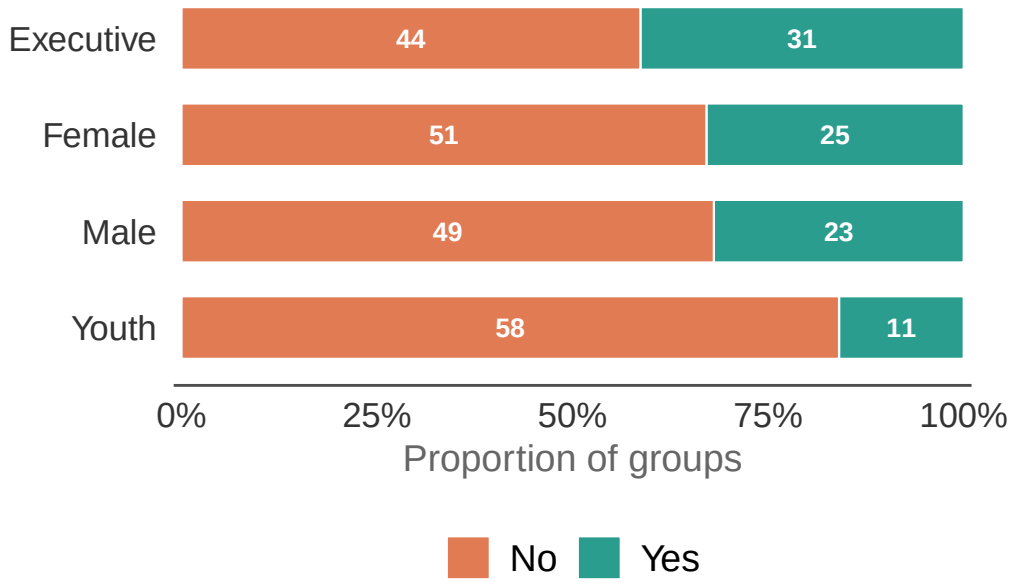

Figure 1: Has CFM improved access to natural resources for household use? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = case_match(
      as.character(economic_transformation_economic_benefits_from_cfm_improved_access_to_natu
      "1" ~ "Yes",
      "0" ~ "No"
    ),
    response = factor(response, levels = c("Yes", "No")),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    )
```

```r
) |>
drop_na(response) |>
count(interview_type, response) |>
group_by(interview_type) |>
mutate(prop = n / sum(n)) |>
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
  guide  = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

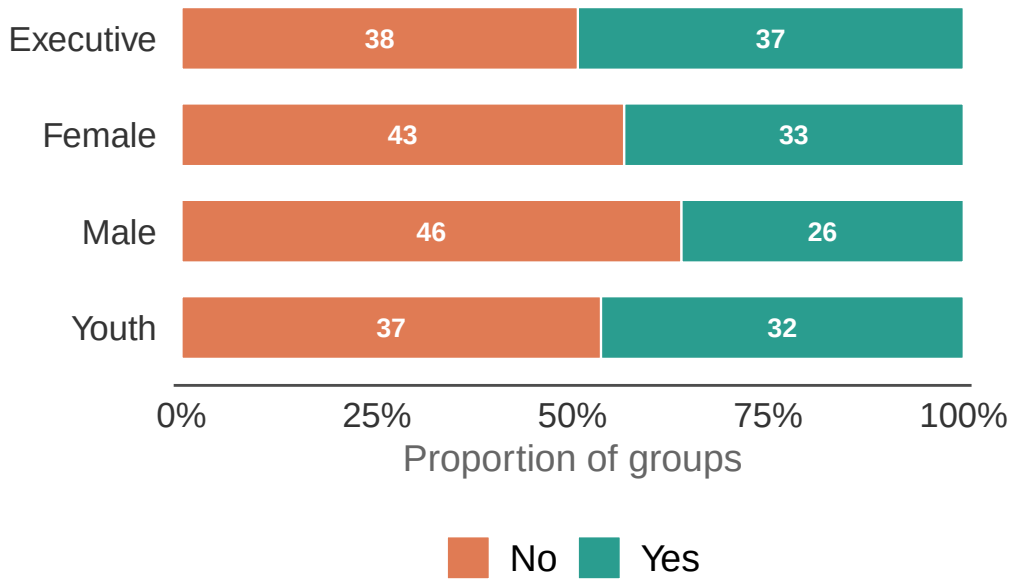

Figure 2: Has CFM improved access to natural resources for sale? The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

Responses were similar for both household use and sale. Most FGD groups said that there was no increase in access to natural resources for household use or for sale. However, approximately one third of the groups indicated that access to natural resources has increased under CFM. Community Youth were significantly more likely to say there had not been any improvement in access to natural resources for household use, as compared to CFMG Executive FGDs. None of the other pairwise comparisons were significant.

> **Access to natural resources - FGDs - Model methods**
>
> To examine differences among FGD types in whether they reported improved access to natural resources as an economic benefit of CFM, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We ran separate models for household use and resources for sale. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
```

```r
    hh = economic_transformation_economic_benefits_from_cfm_improved_access_to_natural_resour
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(hh) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M1 <- glmer(hh ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M1      <- emmeans(M1, specs = ~ interview_type, type = "response")
pairwise.M1 <- pairs(emm.M1, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: hh ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   354.6    376.7   -171.3    342.6      286

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.1578 -0.6593 -0.4742  0.9861  3.2181

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.2708   0.5203
 province      (Intercept) 0.2167   0.4656
Number of obs: 292, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
```

```
(Intercept)           -0.4109     0.3073  -1.337 0.181166
interview_typeFemale  -0.4185     0.3578  -1.170 0.242053
interview_typeMale    -0.4748     0.3656  -1.299 0.194028
interview_typeYouth   -1.4441     0.4253  -3.395 0.000685 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.552
intrvw_typM -0.542  0.483
intrvw_typY -0.449  0.419  0.413


 contrast           odds.ratio    SE  df null z.ratio p.value
 Executive / Female       1.52 0.544 Inf    1   1.170  0.6459
 Executive / Male         1.61 0.588 Inf    1   1.299  0.5637
 Executive / Youth        4.24 1.802 Inf    1   3.395  0.0038
 Female / Male            1.06 0.389 Inf    1   0.153  0.9987
 Female / Youth           2.79 1.188 Inf    1   2.408  0.0756
 Male / Youth             2.64 1.137 Inf    1   2.248  0.1106

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

```r
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    sale = economic_transformation_economic_benefits_from_cfm_improved_access_to_natural_res
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(sale) |>
  mutate(province_cfmg = interaction(province, cfmg_name))
```

```r
M2 <- glmer(sale ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M2      <- emmeans(M2, specs = ~ interview_type, type = "response")
pairwise.M2 <- pairs(emm.M2, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: sale ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   370.5    392.5   -179.2    358.5      286

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.6083 -0.5910 -0.3719  0.6081  2.1084

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 2.1459   1.465
 province      (Intercept) 0.4775   0.691
Number of obs: 292, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)           -0.2141     0.4183  -0.512    0.609
interview_typeFemale  -0.2498     0.3953  -0.632    0.527
interview_typeMale    -0.6337     0.4084  -1.552    0.121
interview_typeYouth   -0.1376     0.4030  -0.341    0.733

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.474
intrvw_typM -0.462  0.497
intrvw_typY -0.458  0.492  0.473

 contrast           odds.ratio    SE  df null z.ratio p.value
 Executive / Female      1.284 0.507 Inf    1   0.632  0.9218
 Executive / Male        1.885 0.770 Inf    1   1.552  0.4066
```

```
 Executive / Youth       1.147 0.462 Inf    1   0.341  0.9863
 Female / Male           1.468 0.592 Inf    1   0.952  0.7768
 Female / Youth          0.894 0.360 Inf    1  -0.279  0.9924
 Male / Youth            0.609 0.254 Inf    1  -1.191  0.6324

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether CFM had improved access to natural resources for household use or for sale. Table 1 shows the percentage reporting each by interviewee role. Again a majority of respondents indicated that they did not think that CFM had improved access to natural resources for either household use or sale. DFOs were significantly more likely to suggest that CFM had improved access to natural resources for household use than CFMG executives or traditional leaders. None of the other pairwise comparisons were significant.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    hh   = economic_transformation_economic_benefits_from_cfm_improved_access_to_natural_reso
    sale = economic_transformation_economic_benefits_from_cfm_improved_access_to_natural_reso
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
      "Community member", "CFMG executive member", "Traditional leader",
      "District Forestry Officer", "NGO representative"
    ))
  ) |>
  group_by(interviewee_role) |>
  summarise(
    n         = n(),
    Household = mean(hh,   na.rm = TRUE),
    Sale      = mean(sale, na.rm = TRUE),
    .groups   = "drop"
```

```r
  ) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(
    interviewee_role = "Role",
    n                = "n",
    Household        = "Household use",
    Sale             = "For sale"
  ) |>
  fmt_percent(
    columns  = c(Household, Sale),
    decimals = 0
  ) |>
  tab_header(
    title    = "Improved access to natural resources from CFM as reported by individual inter
    subtitle = "Values represent the % of responses agreeing that CFM has improved assess to
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "left",   columns = interviewee_role) |>
  cols_align(align = "center", columns = c(n, Household, Sale)) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 1: Percentage of individual interviewees reporting improved access to natural resources as an economic benefit of CFM.

Improved access to natural resources from CFM as reported by individual interviewees

Values represent the % of responses agreeing that CFM has improved assess to resources for household use or sale. n = the total number of interview responses.

Role	n	Household use	For sale
Community member	31	45%	52%
CFMG executive member	63	24%	33%
Traditional leader	75	27%	39%
District Forestry Officer	56	62%	64%
NGO representative	32	50%	38%

Access to natural resources - Individual interviewees - Model methods

To examine differences among individual interviewee roles in whether they reported improved access to natural resources as an economic benefit of CFM, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. We ran separate models for household use and resources for sale. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    hh                 = economic_transformation_economic_benefits_from_cfm_improved_access_t
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(hh) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M4 <- glmer(hh ~ interviewee_role + interviewee_gender + interviewee_age +
              (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M4      <- emmeans(M4, specs = ~ interviewee_role, type = "response")
pairwise.M4 <- pairs(emm.M4, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: hh ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   332.3    364.3   -157.2    314.3      248

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.4353 -0.6697 -0.4973  0.8737  2.2111

Random effects:
```

```
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.1485   0.3854
 province      (Intercept) 0.1294   0.3597
Number of obs: 257, groups:  province_cfmg, 75; province, 8

Fixed effects:
                                 Estimate Std. Error z value Pr(>|z|)
(Intercept)                      0.234974   0.810009   0.290   0.7717
interviewee_rolecfmg_exec_member -1.022963   0.526522  -1.943   0.0520 .
interviewee_roletrad_leader      -0.822630   0.490258  -1.678   0.0934 .
interviewee_roledfo               0.694930   0.502102   1.384   0.1663
interviewee_roleNGO_rep           0.098006   0.555111   0.177   0.8599
interviewee_gendermale           -0.316646   0.371724  -0.852   0.3943
interviewee_age                  -0.002426   0.012651  -0.192   0.8479
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.408
intrvw_rlt_ -0.230  0.592
intrvw_rldf -0.542  0.577  0.583
intrvw_NGO_ -0.538  0.547  0.520  0.593
intrvw_gndr -0.356  0.073 -0.066 -0.045    0.056
interview_g -0.780  0.030 -0.170  0.230    0.228 -0.004

 contrast                       odds.ratio     SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member      2.781 1.4645 Inf    1   1.943  0.2946
 cfmg_member / trad_leader           2.276 1.1161 Inf    1   1.678  0.4476
 cfmg_member / dfo                   0.499 0.2506 Inf    1  -1.384  0.6380
 cfmg_member / NGO_rep               0.907 0.5033 Inf    1  -0.177  0.9998
 cfmg_exec_member / trad_leader      0.818 0.3768 Inf    1  -0.435  0.9926
 cfmg_exec_member / dfo              0.179 0.0850 Inf    1  -3.628  0.0026
 cfmg_exec_member / NGO_rep          0.326 0.1680 Inf    1  -2.175  0.1892
 trad_leader / dfo                   0.219 0.0994 Inf    1  -3.347  0.0073
 trad_leader / NGO_rep               0.398 0.2051 Inf    1  -1.788  0.3807
 dfo / NGO_rep                       1.817 0.8711 Inf    1   1.245  0.7250

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

```r
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    sale               = economic_transformation_economic_benefits_from_cfm_improved_access_t
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(sale) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M5 <- glmer(sale ~ interviewee_role + interviewee_gender + interviewee_age +
              (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M5      <- emmeans(M5, specs = ~ interviewee_role, type = "response")
pairwise.M5 <- pairs(emm.M5, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: sale ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   351.4    383.3   -166.7    333.4      248

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.8650 -0.7390 -0.5522  0.9065  1.7082

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.3328   0.5769
 province      (Intercept) 0.1428   0.3780
Number of obs: 257, groups:  province_cfmg, 75; province, 8

Fixed effects:
                                 Estimate Std. Error z value Pr(>|z|)
(Intercept)                       0.99819    0.81707   1.222    0.222
interviewee_rolecfmg_exec_member -0.62332    0.51852  -1.202    0.229
interviewee_roletrad_leader      -0.46359    0.49225  -0.942    0.346
```

```
interviewee_roledfo               0.49287    0.51928   0.949    0.343
interviewee_roleNGO_rep          -0.76533    0.57648  -1.328    0.184
interviewee_gendermale           -0.44761    0.36476  -1.227    0.220
interviewee_age                  -0.01207    0.01278  -0.944    0.345

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.389
intrvw_rlt_ -0.241  0.606
intrvw_rldf -0.534  0.627  0.603
intrvw_NGO_ -0.531  0.531  0.530  0.581
intrvw_gndr -0.358  0.020 -0.080 -0.036    0.061
interview_g -0.773 -0.011 -0.176  0.200    0.215  0.010


 contrast                       odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member      1.865 0.967 Inf    1   1.202  0.7502
 cfmg_member / trad_leader           1.590 0.783 Inf    1   0.942  0.8806
 cfmg_member / dfo                   0.611 0.317 Inf    1  -0.949  0.8776
 cfmg_member / NGO_rep               2.150 1.239 Inf    1   1.328  0.6739
 cfmg_exec_member / trad_leader      0.852 0.383 Inf    1  -0.355  0.9966
 cfmg_exec_member / dfo              0.328 0.147 Inf    1  -2.492  0.0924
 cfmg_exec_member / NGO_rep          1.153 0.614 Inf    1   0.267  0.9989
 trad_leader / dfo                   0.384 0.173 Inf    1  -2.120  0.2116
 trad_leader / NGO_rep               1.352 0.708 Inf    1   0.576  0.9786
 dfo / NGO_rep                       3.519 1.773 Inf    1   2.497  0.0913

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

Employment

We asked FGD groups whether CFM had generated new employment opportunities (Figure 3).

```
cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    response = economic_transformation_yn_formal_employment_opportunities_created_cfm,
    response = case_match(response,
                          "yes" ~ "Yes",
```

```r
                          "no" ~ "No",
                          .default = NA),
  response = factor(response, levels = c("Yes", "No")),
  interview_type = factor(
    case_match(
      interview_type,
      "cfmg_executive" ~ "Executive",
      "cfmg_female"    ~ "Female",
      "cfmg_male"      ~ "Male",
      "cfmg_youth"     ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female", "Executive")
  )
) |>
drop_na(response) |>
count(interview_type, response) |>
group_by(interview_type) |>
mutate(prop = n / sum(n)) |>
ungroup() |>
ggplot(aes(y = interview_type, x = prop, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_text(
  aes(label = ifelse(prop >= 0.05, n, "")),
  position = position_stack(vjust = 0.5),
  size = 3.5, fontface = "bold", colour = "white"
) +
scale_x_continuous(
  labels = percent_format(accuracy = 1),
  expand = expansion(mult = 0.01),
  name   = "Proportion of groups"
) +
scale_fill_manual(
  values = c("Yes" = "#2A9D8F", "No" = "#E07B54"),
  guide  = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
```

```
    axis.text.y         = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x         = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x        = element_text(size = rel(1), colour = "grey40"),
    legend.position     = "bottom",
    legend.text         = element_text(size = rel(1)),
    plot.margin         = margin(10, 25, 10, 10)
  )
```

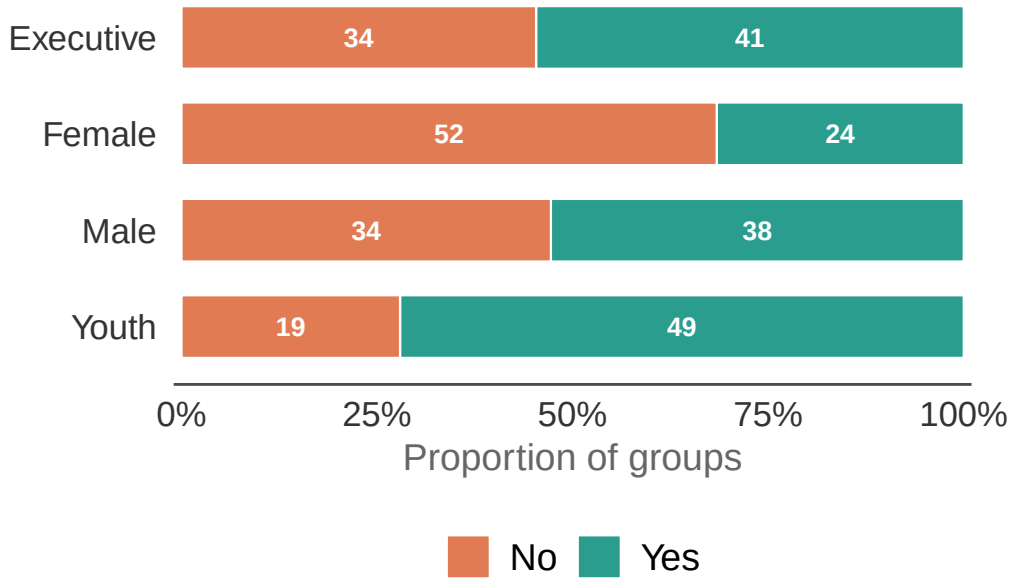

Figure 3: Responses to whether CFM has generated employment opportunities. The length of each bar segment represents the proportion of groups in that category. The numbers superimposed on the bars represent the actual number of groups in each category.

As can be seen from the figure, approximately half of the FGD groups agreed that CFM had increased employment opportunities. Interestingly, community youth were especially likely to agreed that CFM had created employment opportunities. In contrast, community female FGD groups were much more likely to say that CFM had not increased employment opportunities. All pairwise comparisons were highly significant.

Employment - FGDs - Model methods

To examine differences among FGD types in whether they reported CFM had generated employment opportunities, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```r
test.data <- cfmg |>
  filter(interview_type != "individual_interview") |>
  mutate(
    empl = economic_transformation_yn_formal_employment_opportunities_created_cfm,
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M3 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M3      <- emmeans(M3, specs = ~ interview_type, type = "response")
pairwise.M3 <- pairs(emm.M3, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data
```

```
     AIC      BIC   logLik deviance df.resid
   353.8    375.9   -170.9    341.8      285

Scaled residuals:
    Min      1Q  Median      3Q     Max
-3.3885 -0.6761  0.2629  0.6536  2.3311

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.5285   0.727
 province      (Intercept) 1.0731   1.036
Number of obs: 291, groups:  province_cfmg, 75; province, 8

Fixed effects:
                      Estimate Std. Error z value Pr(>|z|)
(Intercept)           0.408529   0.002000   204.3   <2e-16 ***
interview_typeFemale -1.352300   0.002001  -675.9   <2e-16 ***
interview_typeMale   -0.251254   0.002000  -125.6   <2e-16 ***
interview_typeYouth   1.013109   0.002001   506.4   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF 0.000
intrvw_typM 0.000  0.000
intrvw_typY 0.000  0.000  0.000
optimizer (Nelder_Mead) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.0355414 (tol = 0.002, component 1)

 contrast           odds.ratio        SE  df null  z.ratio p.value
 Executive / Female    3.86631 0.0077357 Inf    1  675.881  <.0001
 Executive / Male      1.28564 0.0025712 Inf    1  125.629  <.0001
 Executive / Youth     0.36309 0.0007264 Inf    1 -506.396  <.0001
 Female / Male         0.33252 0.0009407 Inf    1 -389.214  <.0001
 Female / Youth        0.09391 0.0002658 Inf    1 -835.854  <.0001
 Male / Youth          0.28242 0.0007989 Inf    1 -446.949  <.0001

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether CFM had generated employment opportunities (Table 2). Interestingly, community respondents were much more likely to agree that CFM

had created employment opportunities than NGO representatives or DFOs. The differences between DFOs and CFMG executives and traditional leaders were significant, but none of the other pairwise comparisons were significant.

```
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = economic_transformation_yn_formal_employment_opportunities_created_cfm
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
      "Community member", "CFMG executive member", "Traditional leader",
      "District Forestry Officer", "NGO representative"
    )),
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA)

  ) |>
  group_by(interviewee_role) |>
  summarise(
    n          = n(),
    Employment = mean(empl, na.rm = TRUE),
    .groups    = "drop"
  ) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(
    interviewee_role = "Role",
    n                = "n",
    Employment       = "Agree CFM has generated employment opportunities"
  ) |>
  fmt_percent(
```

```
    columns  = Employment,
    decimals = 0
  ) |>
  tab_header(
    title    = "Employment opportunities reported by individual interviewees",
    subtitle = "Values represent the % of responses agreeing that CFM has generated employmen
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "left",   columns = interviewee_role) |>
  cols_align(align = "center", columns = c(n, Employment)) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 2: Percentage of individual interviewees reporting that CFM has generated employment opportunities by interviewee role.

Employment opportunities reported by individual interviewees

Values represent the % of responses agreeing that CFM has generated employment opportunities. n = the total number of interview responses.

Role	n	Agree CFM has generated employment opportunities
Community member	31	60%
CFMG executive member	63	75%
Traditional leader	75	53%
District Forestry Officer	56	38%
NGO representative	32	48%

Employment - Individual interviewees - Model methods

To examine differences among individual interviewee roles in whether they reported CFM had generated employment opportunities, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```r
test.data <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    empl               = economic_transformation_yn_formal_employment_opportunities_created_
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no" ~ 0,
                      .default = NA),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M6 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
              (1 | province) + (1 | province_cfmg),
            data = test.data, family = binomial)

emm.M6      <- emmeans(M6, specs = ~ interviewee_role, type = "response")
pairwise.M6 <- pairs(emm.M6, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data

     AIC      BIC   logLik deviance df.resid
   325.4    357.1   -153.7    307.4      241

Scaled residuals:
    Min      1Q  Median      3Q     Max
-2.5075 -0.8214  0.3940  0.7769  2.0657

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.1954   0.4420
 province      (Intercept) 0.3144   0.5607
Number of obs: 250, groups:  province_cfmg, 73; province, 8

Fixed effects:
```

```
                                Estimate Std. Error z value Pr(>|z|)
(Intercept)                      2.21084    0.92174   2.399  0.01646 *
interviewee_rolecfmg_exec_member 0.33131    0.56397   0.587  0.55690
interviewee_roletrad_leader     -0.08792    0.53129  -0.165  0.86856
interviewee_roledfo             -1.44817    0.57963  -2.498  0.01247 *
interviewee_roleNGO_rep         -0.97536    0.63426  -1.538  0.12410
interviewee_gendermale           0.43584    0.37988   1.147  0.25126
interviewee_age                 -0.04125    0.01396  -2.955  0.00313 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.411
intrvw_rlt_ -0.282  0.643
intrvw_rldf -0.606  0.642  0.644
intrvw_NGO_ -0.573  0.581  0.578  0.661
intrvw_gndr -0.298  0.013 -0.076 -0.019    0.016
interview_g -0.775  0.027 -0.145  0.272    0.262 -0.032
optimizer (Nelder_Mead) convergence code: 0 (OK)
Model failed to converge with max|grad| = 0.00450701 (tol = 0.002, component 1)
```

```
 contrast                       odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member      0.718 0.405 Inf    1  -0.587  0.9770
 cfmg_member / trad_leader           1.092 0.580 Inf    1   0.165  0.9998
 cfmg_member / dfo                   4.255 2.467 Inf    1   2.498  0.0909
 cfmg_member / NGO_rep               2.652 1.682 Inf    1   1.538  0.5377
 cfmg_exec_member / trad_leader      1.521 0.705 Inf    1   0.905  0.8953
 cfmg_exec_member / dfo              5.927 2.868 Inf    1   3.677  0.0022
 cfmg_exec_member / NGO_rep          3.694 2.039 Inf    1   2.367  0.1245
 trad_leader / dfo                   3.897 1.834 Inf    1   2.890  0.0315
 trad_leader / NGO_rep               2.429 1.320 Inf    1   1.633  0.4761
 dfo / NGO_rep                       0.623 0.313 Inf    1  -0.941  0.8809

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

Changes in numbers employed

We asked CFMG Executive FGDs to estimate the number of people employed before and after CFM (Figure 4). The mean number of people employed per CFMG varied from three

in Western Province to 21 in Lusaka Province. The figure also indicates that the numbers of people employed in forest management had substantially increased.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(
    province,
    "Before CFM" = economic_transformation_num_employed_before_cfmg,
    "After CFM"  = economic_transformation_num_employed_after_cfmg
  ) |>
  pivot_longer(-province, names_to = "timing", values_to = "n_employed") |>
  filter(!is.na(n_employed)) |>
  mutate(
    timing     = factor(timing, levels = c("Before CFM", "After CFM")),
    n_employed = if_else(province == "Western" & timing == "After CFM",
                         n_employed / 2, n_employed)
  ) |>
  group_by(province, timing) |>
  summarise(mean_employed = mean(n_employed, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(y = reorder(province, mean_employed), x = mean_employed, fill = timing)) +
  geom_col(
    position = position_dodge(width = 0.7), width = 0.6,
    colour = "white", linewidth = 0.3
  ) +
  geom_text(
    aes(label = round(mean_employed, 0)),
    position = position_dodge(width = 0.7),
    hjust = -0.2, size = 3.5, colour = "grey25", fontface = "bold"
  ) +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.18)),
    name   = "Mean number employed per CFMG"
  ) +
  coord_cartesian(clip = "off") +
  scale_fill_manual(
    values = c("Before CFM" = "#E07B54", "After CFM" = "#2A9D8F")
  ) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 18) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_line(colour = "grey85", linewidth = 0.4),
```

```
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

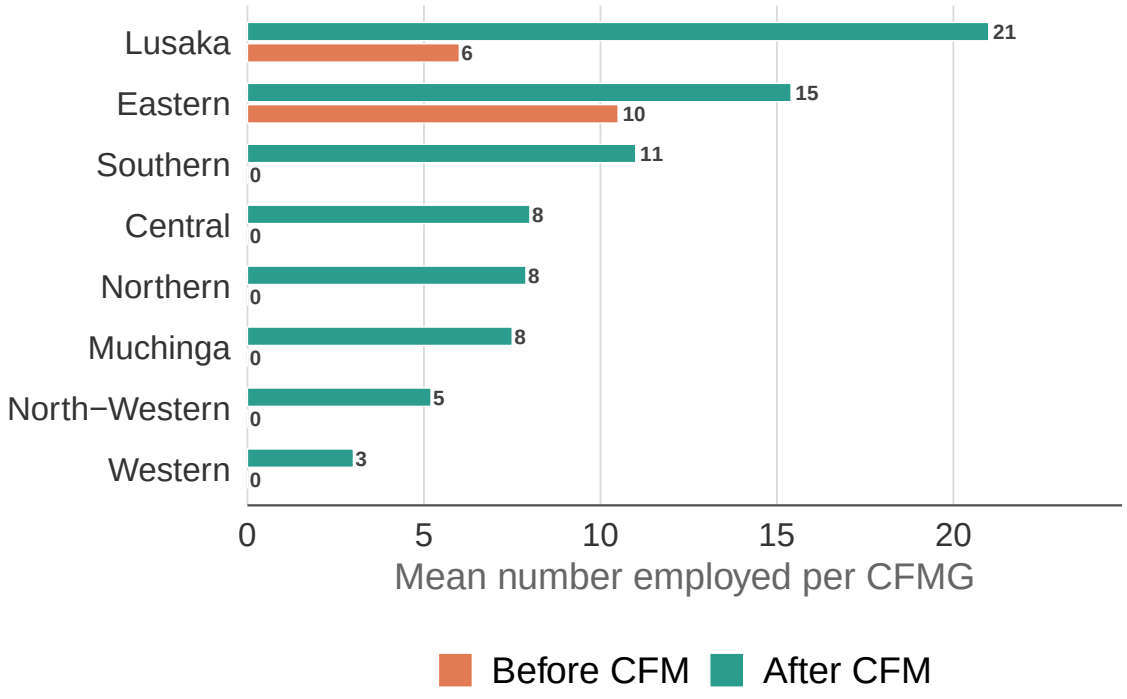

Figure 4: Mean number of people per CFMG employed before and after CFM by province, as reported by CFMG Executives.

Full-time senior employment by gender and province

CFMG Executive FGDs also reported on the number of senior males and females in full-time employment in the CFMG (Figure 5). Clearly, there are many more senior men in full-time employment than women. According to the response across the 75 CFMGs surveyed, 278 senior men were employed as compared to 56 women.

```r
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(
    province,
    "Male"   = economic_transformation_formal_employment_quests_formalemploymentdetails_1__ma
    "Female" = economic_transformation_formal_employment_quests_formalemploymentdetails_1__fe
  ) |>
  pivot_longer(-province, names_to = "gender", values_to = "n_employed") |>
  filter(!is.na(n_employed)) |>
  mutate(gender = factor(gender, levels = c("Male", "Female"))) |>
  group_by(province, gender) |>
  summarise(total = sum(n_employed, na.rm = TRUE), .groups = "drop") |>
  ggplot(aes(y = reorder(province, total), x = total, fill = gender)) +
  geom_col(
    position = position_dodge(width = 0.7), width = 0.6,
    colour = "white", linewidth = 0.3
  ) +
  geom_text(
    aes(label = total),
    position = position_dodge(width = 0.7),
    hjust = -0.2, size = 3.5, colour = "grey25", fontface = "bold"
  ) +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.18)),
    name   = "Number employed full-time"
  ) +
  coord_cartesian(clip = "off") +
  scale_fill_manual(
    values = c("Male" = "#2A9D8F", "Female" = "#E07B54")
  ) +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 18) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_line(colour = "grey85", linewidth = 0.4),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
```

```
    plot.margin         = margin(10, 25, 10, 10)
  )
```

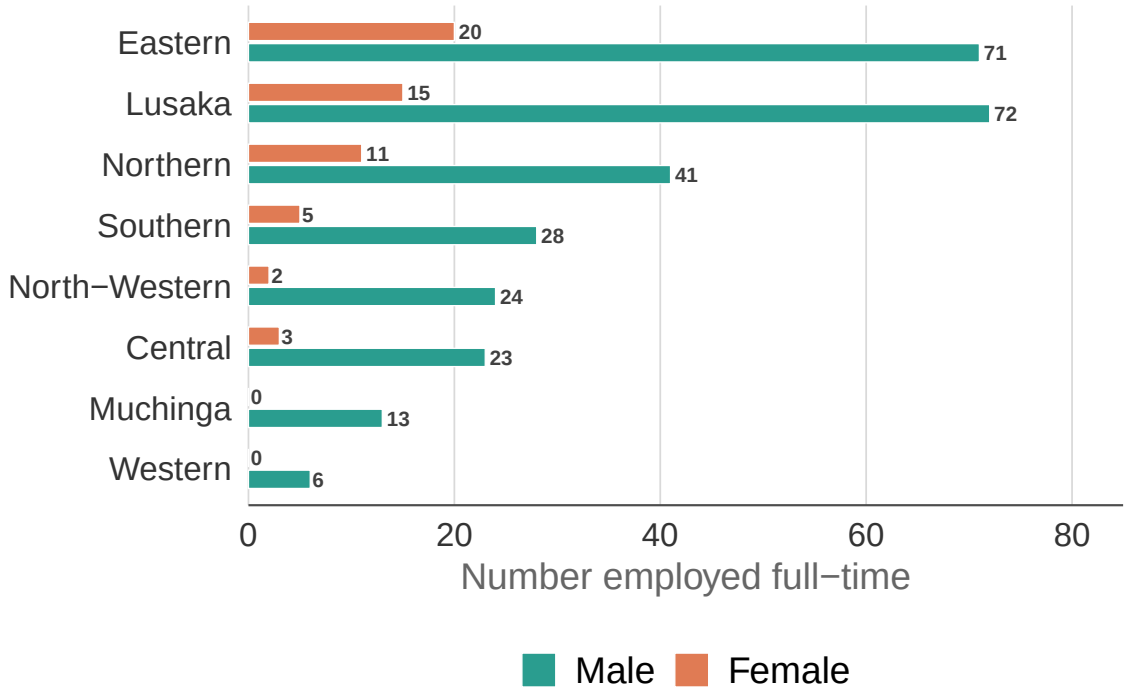

Figure 5: Number of males and females employed full-time in CFMGs, by province, as reported by executive FGDs.

CFMG Executive FGDs also reported the number of male and female youth in full-time employment in the CFMG (Figure 6). Across the sample of 75 CFMGs, 207 male youths were employed comapred to 48 female youths.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(
    province,
    "Male youth"   = economic_transformation_formal_employment_quests_formalemploymentdetails
    "Female youth" = economic_transformation_formal_employment_quests_formalemploymentdetails
  ) |>
  pivot_longer(-province, names_to = "gender", values_to = "n_employed") |>
  filter(!is.na(n_employed)) |>
  mutate(gender = factor(gender, levels = c("Male youth", "Female youth"))) |>
```

```r
group_by(province, gender) |>
summarise(total = sum(n_employed, na.rm = TRUE), .groups = "drop") |>
ggplot(aes(y = reorder(province, total), x = total, fill = gender)) +
geom_col(
  position = position_dodge(width = 0.7), width = 0.6,
  colour = "white", linewidth = 0.3
) +
geom_text(
  aes(label = total),
  position = position_dodge(width = 0.7),
  hjust = -0.2, size = 3.5, colour = "grey25", fontface = "bold"
) +
scale_x_continuous(
  expand = expansion(mult = c(0, 0.18)),
  name   = "Number of youth employed full-time"
) +
coord_cartesian(clip = "off") +
scale_fill_manual(
  values = c("Male youth" = "#2A9D8F", "Female youth" = "#E07B54")
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_line(colour = "grey85", linewidth = 0.4),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
  axis.title.x       = element_text(size = rel(1), colour = "grey40"),
  legend.position    = "bottom",
  legend.text        = element_text(size = rel(1)),
  plot.margin        = margin(10, 25, 10, 10)
)
```

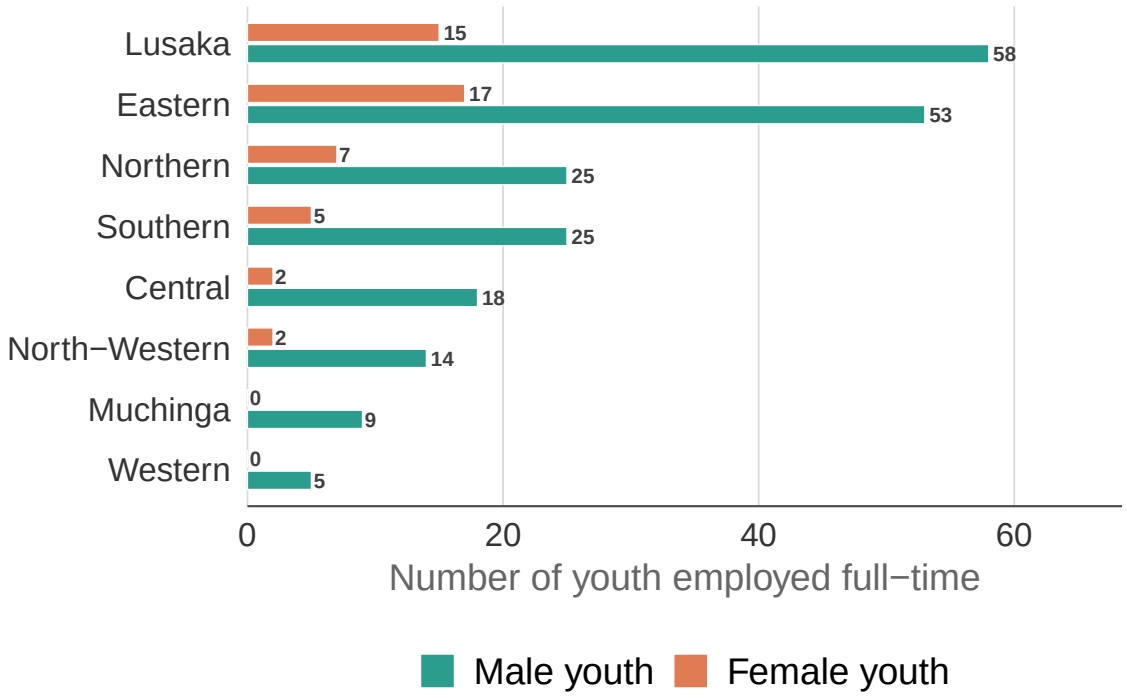

Figure 6: Number of male and female youth employed full-time in CFMGs, by province, as reported by executive FGDs.

Types of formal employment opportunities

Community female, male, and youth FGDs identified the types of formal employment opportunities generated by CFM (Figure 7). Clearly, by far the largest type of job created is Honorary Forest Office or Community Scout.

```
prefix <- "economic_transformation_formal_employment_opportunities_from_cfm_"

job_labels <- c(
  "forest_managers"                                          = "Forest managers",
  "forest_rangers"                                           = "Forest rangers",
  "community_scouts"                                         = "Community scouts",
  "hfos"                                                     = "HFOs",
  "monitoring_and_evaluation_officers"                       = "M&E officers",
  "administrative_staff"                                     = "Admin staff",
  "program_coordinators"                                     = "Program coordinators",
  "training_coordinators_facilitators"                       = "Training coordinators",
  "agroforestry_extension_officers"                          = "Agroforestry extension",
```

```
  "nursery_managers"                                          = "Nursery managers",
  "nursery_workers"                                           = "Nursery workers",
  "technical_support_roles_e_g_soil_scientists_agronomists"  = "Technical support",
  "eco_tourism_operators_e_g_tour_guides_lodge_staff"         = "Eco-tourism operators",
  "cultural_heritage_and_craft_trainers"                      = "Cultural heritage & craft",
  "craftsmen_and_woodcarvers"                                 = "Craftsmen & woodcarvers",
  "processing_plant_staff_e_g_honey_timber_charcoal"          = "Processing natural resources"
  "marketing_and_sales_agents"                                = "Marketing & sales",
  "researchers"                                               = "Researchers",
  "data_analysts"                                             = "Data analysts",
  "surveyors"                                                 = "Surveyors",
  "field_data_collectors"                                     = "Field data collectors",
  "biodiversity_monitors"                                     = "Biodiversity monitors",
  "forest_extension_workers"                                  = "Forest extension workers",
  "forest_protection_and_fire_mnagement"                      = "Forest protection & fire",
  "enterprise_and_processing_roles"                           = "Enterprise",
  "training_and_facilitation"                                 = "Training & facilitation",
  "security_services"                                         = "Security services",
  "livestock_and_resource_management"                         = "Livestock & resource mgmt",
  "interns_trainees"                                          = "Interns & trainees",
  "technical_extension_roles"                                 = "Technical extension"
)

full_var_names <- paste0(prefix, names(job_labels))

binary_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_var_names)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = job_labels[sub(prefix, "", variable)]) |>
  filter(freq > 0) |>
  select(word, freq)

comment_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(text = economic_transformation_formal_employment_opportunities_from_cfm_comment) |>
  filter(!is.na(text), trimws(text) != "") |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = "word") |>
  mutate(word = case_match(word,
    "hammer" ~ "Hammer mill",
```

```r
    "mill"   ~ "Hammer mill",
    "cows"   ~ "Cattle",
    .default = word
  )) |>
  count(word, name = "freq")

wc_data <- bind_rows(binary_freq, comment_freq) |>
  group_by(word) |>
  summarise(freq = sum(freq), .groups = "drop")

set.seed(42)
ggplot(wc_data, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 30)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 7: Types of formal employment opportunities reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting that job type.

Income earning opportunities

Carbon finance

We asked the CFMG Executive FGDs whether their CFMG had received income from carbon finance (Figure 8). According to the responses, around one third of CFs have received carbon finance. Later responses suggest this is an under-representation for the number of CFMGs involved in carbon finance and probably reflects only those CFMGs that have actually started to receive payouts from carbon.

```
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  transmute(
    province,
    response = case_when(
```

```
    economic_transformation_cfma_involved_carbon_finance == "yes"
    str_starts(economic_transformation_cfma_involved_carbon_finance, "Yes")
    economic_transformation_cfma_involved_carbon_finance == "no"
    str_starts(economic_transformation_cfma_involved_carbon_finance, "No")
    economic_transformation_cfma_involved_carbon_finance %in% c(
      "Not yet", "At the moment_they are not",
      "The cfmg is yet to enter into partnership with the carbon credit organization",
      "The CFMG has not signed form 7 and MOU. They had to cancel the one they had signed e
      "Has not signed the agreement with CCS yet due to dispute  over percentages in the be
    ) ~ "No",
    TRUE ~ NA_character_
  )
) |>
drop_na(response) |>
mutate(response = factor(response, levels = c("Yes", "No"))) |>
count(province, response) |>
mutate(n_signed = if_else(response == "No", -n, n)) |>
ggplot(aes(y = reorder(province, n_signed), x = n_signed, fill = response)) +
geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
geom_text(
  aes(label  = n,
      hjust  = if_else(response == "No", -0.4, -0.4),
      colour = response),
  size = 4, fontface = "bold"
) +
scale_colour_manual(values = c("No" = "white", "Yes" = "grey25"),
                    guide = "none") +
scale_fill_manual(values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
                  guide  = guide_legend(reverse = TRUE)) +
scale_x_continuous(labels = ~abs(.x),
                   expand = expansion(add = 1.5),
                   name   = "Number of CFMGs") +
scale_y_discrete(expand = expansion(add = 0.5)) +
coord_cartesian(clip = "off") +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
```

```
    axis.text           = element_text(colour = "grey20"),
    legend.position     = "bottom",
    plot.margin         = margin(10, 25, 10, 10)
  )
```

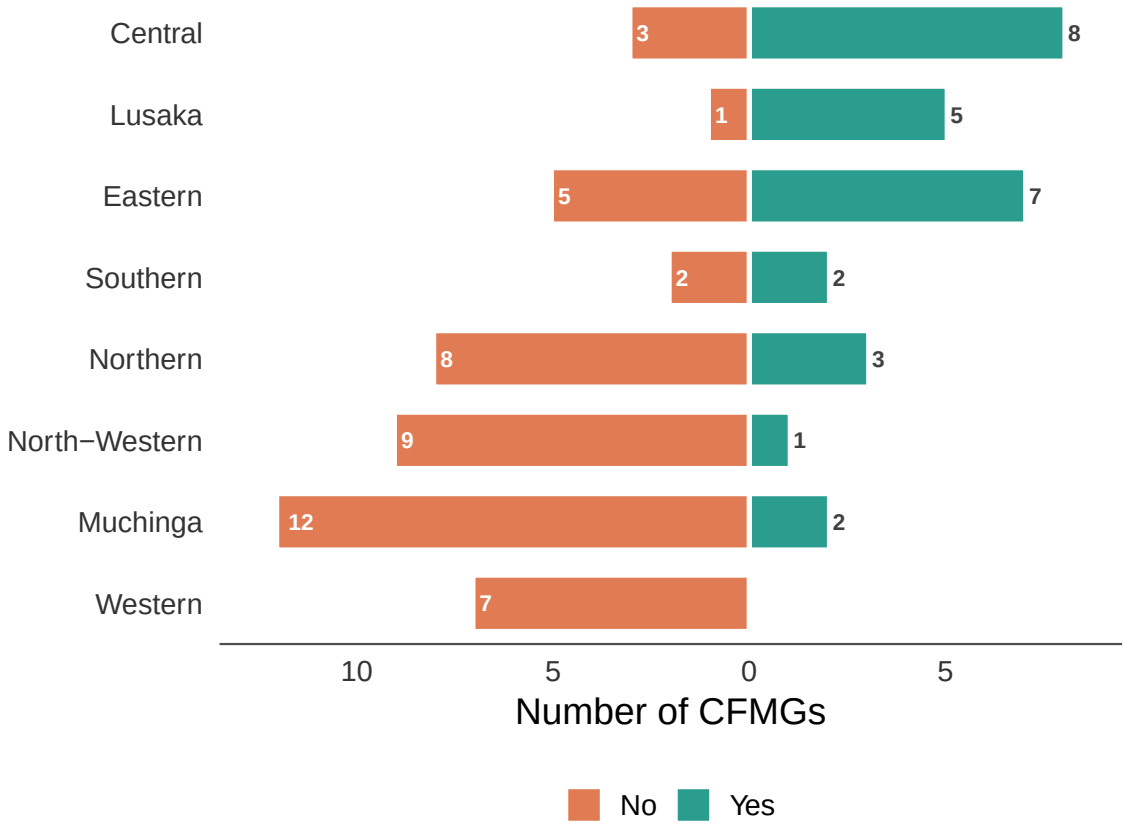

Figure 8: CFMG involvement in carbon finance by province, as reported by CFMG executives.

FGD groups were also asked whether the community was aware of the carbon credit agreement (Figure 9). While most CFMG Executive groups stated that the community was aware of the carbon finance agreement, interestingly two stated that they were not, which is a little surprising. Most community female and male groups indicated they were aware of the carbon finance agreement, but more than half of the community youth groups stated they were not aware of the carbon finance agreement. Community youth were significantly less likely to know about the carbon finance agreement than CFMG Executives, but other pairwise comparisons were not significant.

```r
cfmg |>
  filter(interview_type %in% c("cfmg_executive", "cfmg_female",
                               "cfmg_male", "cfmg_youth")) |>
  transmute(
    group = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Youth", "Male", "Female", "Executive")
    ),
    response = case_when(
      economic_transformation_community_awareness_carbon_credit_agreement == "yes" ~ "Yes",
      economic_transformation_community_awareness_carbon_credit_agreement == "no"  ~ "No",
      economic_transformation_community_awareness_carbon_credit_agreement %in% c(
        "Not aware of the agreement of the Juristiction project we just know what's involved"
        "The agreement is not clear we don't know where they sale the carbon and we don't kno
        "So far we have just heard the chief signed"
      ) ~ "No",
      TRUE ~ NA_character_
    )
  ) |>
  drop_na(response) |>
  mutate(response = factor(response, levels = c("Yes", "No"))) |>
  count(group, response) |>
  mutate(n_signed = if_else(response == "No", -n, n)) |>
  ggplot(aes(y = group, x = n_signed, fill = response)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
  geom_text(
    aes(label  = n,
        hjust  = if_else(response == "No", -0.4, -0.4),
        colour = response),
    size = 4, fontface = "bold"
  ) +
  scale_colour_manual(values = c("No" = "white", "Yes" = "grey25"),
                      guide = "none") +
  scale_fill_manual(values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
                    guide  = guide_legend(reverse = TRUE)) +
```

```r
scale_x_continuous(labels = ~abs(.x),
                   expand = expansion(add = 3),
                   name   = "Number of FGDs") +
scale_y_discrete(expand = expansion(add = 0.5)) +
coord_cartesian(clip = "off") +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text          = element_text(colour = "grey20"),
  legend.position    = "bottom",
  plot.margin        = margin(10, 25, 10, 10)
)
```

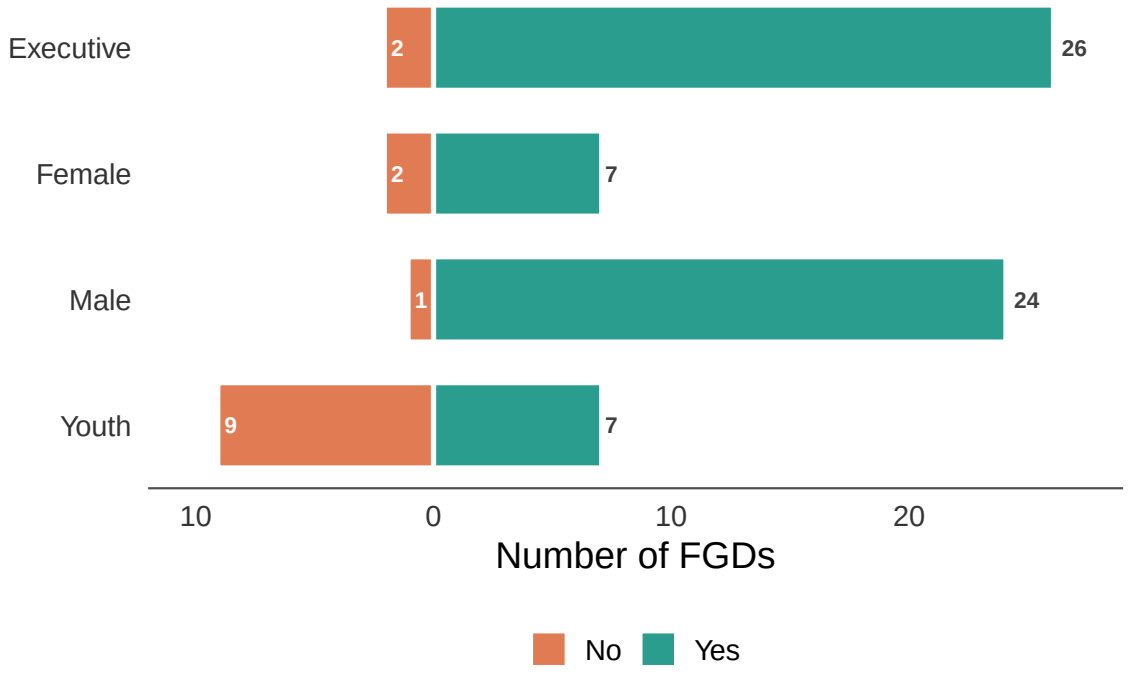

Figure 9: Community awareness of the carbon credit agreement by FGD group.

Carbon finance awareness – FGDs – Model methods

To examine differences among FGD types in whether they reported community awareness of the carbon credit agreement, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```
test.data.ca <- cfmg |>
  filter(interview_type %in% c("cfmg_executive", "cfmg_female",
                               "cfmg_male", "cfmg_youth")) |>
  transmute(
    province, cfmg_name,
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    ),
    empl = case_when(
      economic_transformation_community_awareness_carbon_credit_agreement == "yes" ~ 1L,
      economic_transformation_community_awareness_carbon_credit_agreement == "no"  ~ 0L,
      economic_transformation_community_awareness_carbon_credit_agreement %in% c(
        "Not aware of the agreement of the Juristiction project we just know what's involved
        "The agreement is not clear we don't know where they sale the carbon and we don't kn
        "So far we have just heard the chief signed"
      ) ~ 0L,
      TRUE ~ NA_integer_
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M11 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
             data = test.data.ca, family = binomial)

emm.M11      <- emmeans(M11, specs = ~ interview_type, type = "response")
pairwise.M11 <- pairs(emm.M11, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data.ca

     AIC      BIC   logLik deviance df.resid
    66.3     80.4    -27.1     54.3       72

Scaled residuals:
    Min      1Q  Median      3Q     Max
-4.8990  0.2041  0.2774  0.2774  1.1339

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0        0
 province      (Intercept) 0        0
Number of obs: 78, groups:  province_cfmg, 35; province, 7

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)            2.5649     0.7338   3.495 0.000473 ***
interview_typeFemale  -1.3122     1.0869  -1.207 0.227320
interview_typeMale     0.6131     1.2570   0.488 0.625733
interview_typeYouth   -2.8163     0.8902  -3.164 0.001558 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.675
intrvw_typM -0.584  0.394
intrvw_typY -0.824  0.557  0.481
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

 contrast           odds.ratio     SE  df null z.ratio p.value
 Executive / Female      3.714  4.037 Inf    1   1.207  0.6222
 Executive / Male        0.542  0.681 Inf    1  -0.488  0.9618
 Executive / Youth      16.714 14.879 Inf    1   3.164  0.0085
 Female / Male           0.146  0.189 Inf    1  -1.483  0.4475
 Female / Youth          4.500  4.262 Inf    1   1.588  0.3853
```

```
 Male / Youth           30.857 35.123 Inf    1   3.013  0.0138

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether the community was aware of the carbon credit agreement (Table 3). Most respondents indicated that the community was aware, but the percentage of positive responses varied across interviewee roles. Interestingly, the NGO represenatatives (a groups that also included carbon enterprise representatives), overwhelmingly (~95%) suggested the community was informed, but the only 70% of DFOs thought the communities were informed. However, none of these differences were significant.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = economic_transformation_community_awareness_carbon_credit_agreement
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
      "Community member", "CFMG executive member", "Traditional leader",
      "District Forestry Officer", "NGO representative"
    )),
    empl = case_match(empl,
                      "yes" ~ 1,
                      "no"  ~ 0,
                      .default = NA)
  ) |>
  group_by(interviewee_role) |>
  summarise(
    n         = n(),
    Awareness = mean(empl, na.rm = TRUE),
    .groups   = "drop"
  ) |>
```

```r
arrange(interviewee_role) |>
gt() |>
cols_label(
  interviewee_role = "Role",
  n                = "n",
  Awareness        = "Aware of carbon credit agreement"
) |>
fmt_percent(
  columns  = Awareness,
  decimals = 0
) |>
tab_header(
  title    = "Community awareness of the carbon credit agreement, as reported by individual
  subtitle = "Values represent the % of positive responses. n = the total number of interv
) |>
opt_align_table_header(align = "left") |>
cols_align(align = "left",   columns = interviewee_role) |>
cols_align(align = "center", columns = c(n, Awareness)) |>
tab_style(
  style     = cell_text(align = "center"),
  locations = cells_column_labels(columns = interviewee_role)
) |>
sub_missing()
```

Table 3: Percentage of individual interviewees reporting community awareness of the carbon credit agreement, by interviewee role.

Community awareness of the carbon credit agreement, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses.

Role	n	Aware of carbon credit agreement
Community member	31	83%
CFMG executive member	63	87%
Traditional leader	75	80%
District Forestry Officer	56	70%
NGO representative	32	95%

Carbon finance awareness – Individual interviewees – Model methods

To examine differences among individual interviewee roles in whether they reported community awareness of the carbon credit agreement, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data.ca.ind <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
    empl = case_match(
      economic_transformation_community_awareness_carbon_credit_agreement,
      "yes" ~ 1,
      "no"  ~ 0,
      .default = NA
    ),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M12 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
              (1 | province) + (1 | province_cfmg),
            data = test.data.ca.ind, family = binomial)

emm.M12      <- emmeans(M12, specs = ~ interviewee_role, type = "response")
pairwise.M12 <- pairs(emm.M12, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data.ca.ind

     AIC      BIC   logLik deviance df.resid
    93.3    116.4    -37.6     75.3       88

Scaled residuals:
```

```
    Min      1Q  Median      3Q     Max
-3.3508  0.0000  0.3754  0.4964  0.7812

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0        0
 province      (Intercept) 0        0
Number of obs: 97, groups:  province_cfmg, 41; province, 8

Fixed effects:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                       19.33340 6639.25770   0.003    0.998
interviewee_rolecfmg_exec_member   0.30550    0.98077   0.311    0.755
interviewee_roletrad_leader       -0.03805    1.07042  -0.036    0.972
interviewee_roledfo               -0.45602    0.98520  -0.463    0.643
interviewee_roleNGO_rep            1.64453    1.33587   1.231    0.218
interviewee_gendermale           -19.02900 6639.25756  -0.003    0.998
interviewee_age                    0.01956    0.02862   0.684    0.494

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__  0.000
intrvw_rlt_  0.000  0.643
intrvw_rldf  0.000  0.654  0.556
intrvw_NGO_  0.000  0.482  0.408  0.543
intrvw_gndr -1.000  0.000  0.000  0.000    0.000
interview_g  0.000 -0.084 -0.255  0.248    0.189  0.000
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')

 contrast                        odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member      0.737 0.723 Inf    1  -0.311  0.9980
 cfmg_member / trad_leader           1.039 1.112 Inf    1   0.036  1.0000
 cfmg_member / dfo                   1.578 1.554 Inf    1   0.463  0.9906
 cfmg_member / NGO_rep               0.193 0.258 Inf    1  -1.231  0.7332
 cfmg_exec_member / trad_leader      1.410 1.227 Inf    1   0.395  0.9949
 cfmg_exec_member / dfo              2.142 1.750 Inf    1   0.932  0.8846
 cfmg_exec_member / NGO_rep          0.262 0.319 Inf    1  -1.100  0.8069
 trad_leader / dfo                   1.519 1.476 Inf    1   0.430  0.9929
 trad_leader / NGO_rep               0.186 0.247 Inf    1  -1.267  0.7114
 dfo / NGO_rep                       0.122 0.141 Inf    1  -1.824  0.3596
```

```
Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

Carbon trader support

The types of CFM activities supported by the carbon trader are shown in Figure 10. As can be seen support focused on aspects of forest management or agricultural extension. Other potential elements of support, such as NTFPs, infrastructure development or market access, featured substantially less.

```r
prefix_ts <- "economic_transformation_cfm_activities_supported_by_carbon_trader_"

ts_labels <- c(
  "patrolling_and_monitoring"                        = "Patrolling",
  "fire_management"                                  = "Fire management",
  "biodiversity_conservation"                        = "Biodiversity",
  "agroforestry"                                     = "Agroforestry",
  "climate_smart_farming"                            = "Climate-smart farming",
  "non_timber_forest_products_ntfps"                 = "NTFPs",
  "eco_tourism"                                      = "Eco-tourism",
  "tree_planting"                                    = "Tree planting",
  "training_and_capacity_building"                   = "Training",
  "carbon_stock_tracking"                            = "Carbon tracking",
  "market"                                           = "Market access",
  "alternative_livelihood_activities"                = "Livelihoods",
  "infractrure"                                      = "Infrastructure",
  "social_gatherings"                                = "Social gatherings",
  "employment"                                       = "Employment",
  "community_sensitization_awareness_engangement"    = "Sensitisation"
)

full_ts_vars <- paste0(prefix_ts, names(ts_labels))

ts_freq <- cfmg |>
  select(all_of(full_ts_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = ts_labels[sub(prefix_ts, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)
```

```r
set.seed(42)
ggplot(ts_freq, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 18)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 10: Types of CFM activities supported by the carbon trader.

Other income earning opportunities

FGD groups also reported whether CFM had generated income earning opportunities through specific interventions, including market linkages, alternative livelihood interventions, conservation revenue, forest product processing and ecotourism (Figure 11). Although some community FGDs indicated improvement in income earning opportunities under some categories, especially conservation revenue and forest product processing, most indicated that CFM had not generated new income earning opportunities in these areas.

```r
cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  mutate(interview_type = case_match(
    interview_type,
    "cfmg_female" ~ "Female",
    "cfmg_male"   ~ "Male",
    "cfmg_youth"  ~ "Youth"
  )) |>
  select(
    interview_type,
    "Ecotourism income"         = economic_transformation_economic_benefits_from_cfm_income_
    "Forest product processing" = economic_transformation_economic_benefits_from_cfm_income_
    "Conservation revenue"      = economic_transformation_economic_benefits_from_cfm_revenue_
    "Alternative livelihoods"   = `economic_transformation_economic_benefits_from_cfm_alterna
    "Market linkages"           = economic_transformation_economic_benefits_from_cfm_market_
  ) |>
  pivot_longer(-interview_type, names_to = "income_source", values_to = "value") |>
  filter(!is.na(value)) |>
  mutate(
    response = case_match(as.character(value), "1" ~ "Yes", "0" ~ "No"),
    response = factor(response, levels = c("No", "Yes")),
    income_source = factor(income_source, levels = c(
      "Ecotourism income", "Forest product processing", "Conservation revenue",
      "Alternative livelihoods", "Market linkages"
    )),
    interview_type = factor(interview_type, levels = c("Female", "Male", "Youth"))
  ) |>
  count(interview_type, income_source, response) |>
  mutate(n_signed = if_else(response == "No", -n, n)) |>
  (\(d) ggplot(d, aes(y = income_source, x = n_signed, fill = response)) +
    geom_blank(data = tibble(
      interview_type = factor(
        c("Female", "Female", "Male", "Male", "Youth", "Youth"),
        levels = c("Female", "Male", "Youth")
      ),
      income_source = factor(
        rep("Ecotourism income", 6),
        levels = c("Ecotourism income", "Forest product processing",
                   "Conservation revenue", "Alternative livelihoods", "Market linkages")
      ),
      n_signed = c(-60, 0, -60, 10, -60, 10),
      response = factor(c("No", "Yes", "No", "Yes", "No", "Yes"), levels = c("No", "Yes"))
```

```r
  )) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
  geom_text(
    aes(label  = n,
        hjust  = if_else(response == "No", -0.4, -0.4),
        colour = response),
    size = 3, fontface = "bold"
  ) +
  scale_colour_manual(
    values = c("No" = "white", "Yes" = "grey25"),
    guide  = "none"
  ) +
  scale_x_continuous(
    labels = ~abs(.x),
    breaks = seq(-60, 10, by = 20),
    expand = expansion(add = 0),
    name   = "Number of FGDs"
  ) +
  coord_cartesian(clip = "off") +
  scale_y_discrete(expand = expansion(add = 0.5)) +
  scale_fill_manual(
    values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
    guide  = guide_legend(reverse = FALSE)
  ) +
  facet_wrap(~ interview_type, scales = "free_x") +
  labs(y = NULL, fill = NULL) +
  theme_minimal(base_size = 18) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    strip.text         = element_text(size = rel(1.14), face = "bold"),
    plot.margin        = margin(10, 25, 10, 10)
  )
)()
```

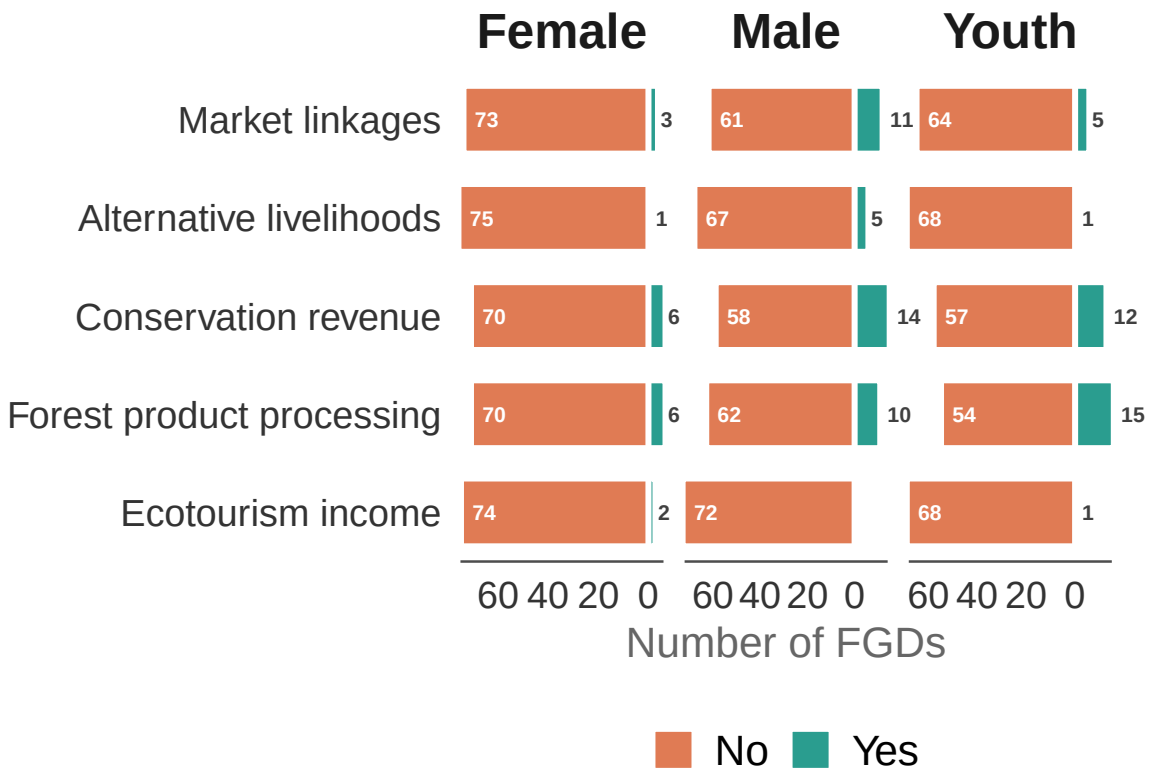

Figure 11: Income earning opportunities generated by CFM, as reported by female, male, and youth FGDs. The length of the bars represents the number of FGDs reporting a benefit (Yes) or no benefit (No) for each income source.

However, community female, male, and youth FGD groups did identified other additional sources of income generated by CFM (Figure 12). Natural resource based value chains featured prominently among these, but interestingly community business was also mentioned.

```
prefix_income <- "economic_transformatioNAdditional_sources_of_income_additional_income_sourc

income_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(starts_with(prefix_income)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(
    word = variable |>
      sub(prefix_income, "", x = _, fixed = TRUE) |>
```

```r
      gsub("_", " ", x = _) |>
      str_to_title(),
    word = case_match(word,
      "Community Businesses And Enterprises" ~ "Community business",
      "Wildlife Based And Safari Income"  ~ "Wildlife based",
      .default = word
    )
  ) |>
  filter(
    freq > 0,
    !str_starts(word, "Rank"),
    !word %in% c("None", "Dont Know")
  ) |>
  select(word, freq)

set.seed(42)
ggplot(income_freq, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 12: Additional sources of income from CFM reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each income source.

To better understand the importance of different income earning opportunities we asked the community female, male, and youth FGD groups to rank their top three additional sources of income from CFM (Figure 13). Interestingly, Among the 14 value chains identified, 12 were mentioned as 1st rank at least once. This immediately points to the diversity of forest products being harvested from CFs across the country, but also to the fact that for most CFs only one product features strongly as an addition source of income. Among these, mushrooms and honey are clearly the most frequently mentioned.

```
cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(matches("Additional_sources_of_income.*rank_")) |>
  pivot_longer(everything(), names_to = "variable", values_to = "rank_val") |>
  mutate(
    rank_val = as.numeric(rank_val),
    source   = sub(".*rank_", "", variable) |>
```

```r
    gsub("_", " ", x = _) |>
    str_to_title(),
  source = case_match(source,
    "Wildlife Based And Safari Income"  ~ "Wildlife Based",
    "Community Businesses And Enterprises" ~ "Community Business",
    .default = source
  )
) |>
filter(rank_val %in% c(1, 2, 3), source != "Carbon Credits") |>
mutate(rank = factor(paste0("Rank ", rank_val), levels = c("Rank 3", "Rank 2", "Rank 1")))
count(source, rank) |>
group_by(source) |>
mutate(total = sum(n)) |>
ungroup() |>
ggplot(aes(y = reorder(source, total), x = n, fill = rank)) +
geom_col(width = 0.7, colour = "white", linewidth = 0.3) +
geom_text(
  aes(label = n),
  position = position_stack(vjust = 0.5),
  size = 3.5, colour = "white", fontface = "bold"
) +
scale_x_continuous(
  expand = expansion(mult = c(0, 0.05)),
  name   = "Number of FGDs"
) +
scale_fill_manual(
  values = c(
    "Rank 1" = "#264653",
    "Rank 2" = "#E9C46A",
    "Rank 3" = "#E07B54"
  ),
  guide = guide_legend(reverse = TRUE)
) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_line(colour = "grey85", linewidth = 0.4),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
  axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
```

```
    axis.title.x        = element_text(size = rel(1), colour = "grey40"),
    legend.position     = "bottom",
    legend.text         = element_text(size = rel(1)),
    plot.margin         = margin(10, 25, 10, 10)
  )
```

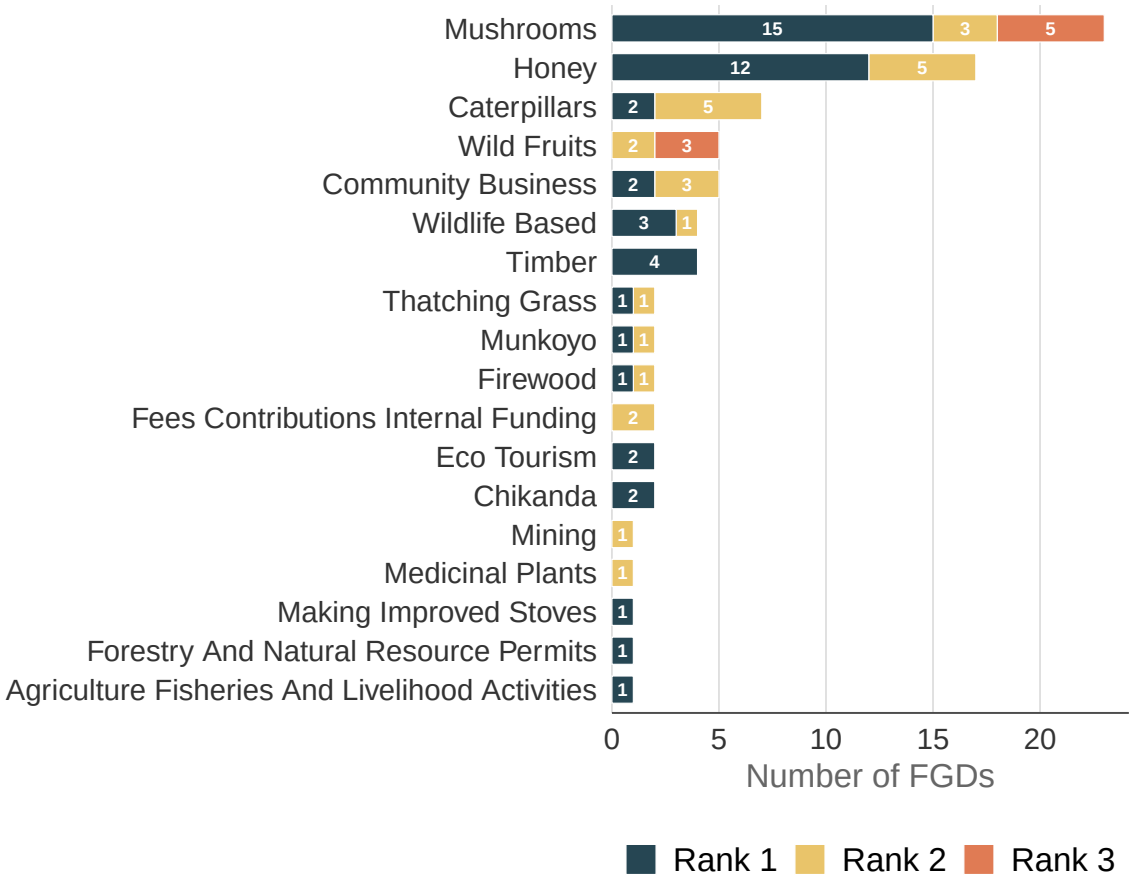

Figure 13: Ranking of additional income sources from CFM by female, male, and youth FGDs. Bars show the number of FGDs assigning each rank. Only income sources receiving at least one top-three ranking are shown.

CFMG Executive FGDs were asked the estimated annual income generated through CFM (Figure 14). Unfortunately, there was a lot of inconsistency in the responses to this question. Most FGD groups did not include income from carbon credit sales, but some did. The figure below includes only those that did not include the carbon credit income to facilitate comparison.

```r
cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(
    province,
    income = economic_transformation_estimated_annual_income_from_cfm
  ) |>
  filter(!is.na(income), income <= 250000) |>
  mutate(province = reorder(province, income, FUN = median)) |>
  ggplot(aes(y = province, x = income)) +
  geom_boxplot(
    width = 0.5, fill = "grey92", colour = "grey40",
    outlier.shape = NA
  ) +
  geom_jitter(
    height = 0.15, width = 0,
    size = 2, alpha = 0.7, colour = "#2A9D8F"
  ) +
  stat_summary(
    fun = mean, geom = "point",
    colour = "red", size = 3, shape = 16
  ) +
  scale_x_continuous(
    labels = label_comma(),
    expand = expansion(mult = c(0.02, 0.1)),
    name   = "Estimated annual income from CFM"
  ) +
  labs(y = NULL) +
  theme_minimal(base_size = 18) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_line(colour = "grey85", linewidth = 0.4),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

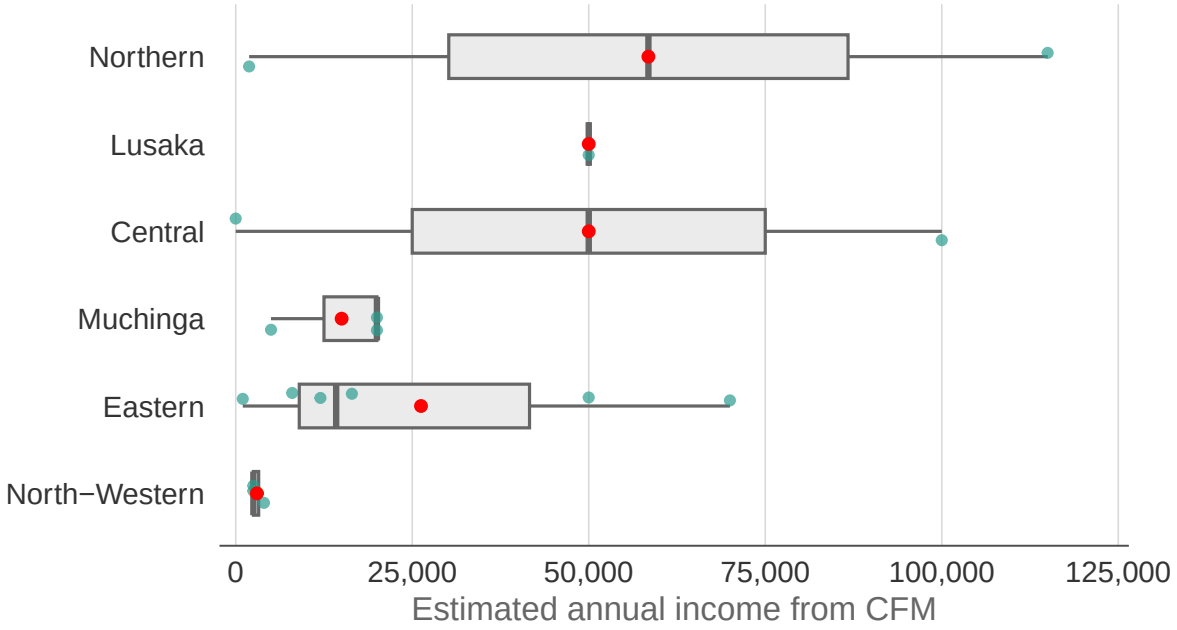

Figure 14: Mean estimated annual income from CFM by province, as reported by executive FGDs.

Female, male, and youth FGDs also reported which forest products were harvested for household use (Figure 15). Mushrooms again featured prominently, but other natural resources are also clearly important, such as wild fruits, medicinal plants, thatching grass and caterpillars.

```
prefix_hh <- "economic_transformation_products_harvested_household_use_"

hh_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(starts_with(prefix_hh)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(
    word = variable |>
      sub(prefix_hh, "", x = _, fixed = TRUE) |>
      gsub("_", " ", x = _) |>
      str_to_title()
  ) |>
  filter(
    freq > 0,
    !word %in% c("Carbon Credits", "Dont Know", "None")
```

```
  ) |>
  select(word, freq)

set.seed(42)
ggplot(hh_freq, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

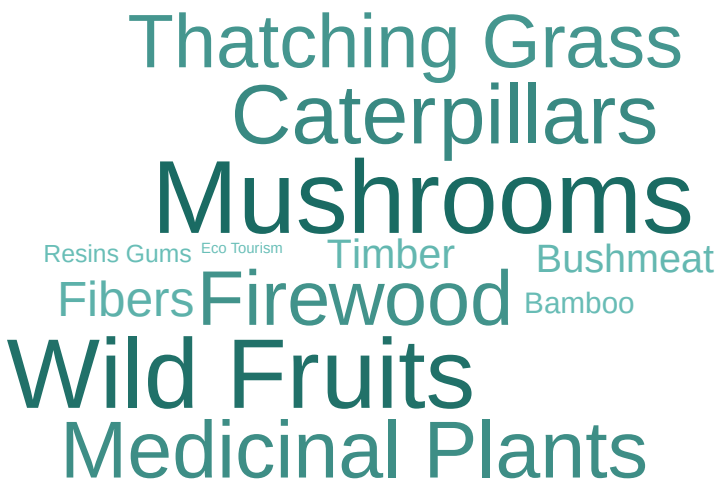

Figure 15: Forest products harvested for household use reported by female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each product.

CFMG Executive FGDs were asked whether the CFMG was collaborating with any private enterprises (Figure 16). The answers indicate that only a small number of CFMGs were collaborating with a private enterprise.

```r
collab_theme <- list(
  theme_minimal(base_size = 18),
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text.y        = element_text(size = rel(1.14), colour = "grey20"),
    axis.text.x        = element_text(size = rel(1.14), colour = "grey20"),
    axis.title.x       = element_text(size = rel(1), colour = "grey40"),
    legend.position    = "bottom",
    legend.text        = element_text(size = rel(1)),
    plot.title         = element_text(size = rel(1.14), face = "bold"),
    plot.margin        = margin(10, 25, 10, 10)
  )
)

# Plot 1: private enterprise collaboration (Yes / No)
p1 <- cfmg |>
  filter(interview_type == "cfmg_executive") |>
  transmute(
    province,
    response = case_match(
      economic_transformation_cfmg_private_enterprise_collaboration,
      "yes" ~ "Yes", "no" ~ "No"
    )
  ) |>
  drop_na(response) |>
  mutate(response = factor(response, levels = c("Yes", "No"))) |>
  count(province, response) |>
  mutate(n_signed = if_else(response == "No", -n, n)) |>
  ggplot(aes(y = reorder(province, n_signed), x = n_signed, fill = response)) +
  geom_col(width = 0.65, colour = "white", linewidth = 0.4) +
  geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
  geom_text(
    aes(label  = n,
        hjust  = if_else(response == "No", -0.4, -0.4),
        colour = response),
    size = 3.5, fontface = "bold"
  ) +
  scale_colour_manual(values = c("No" = "white", "Yes" = "grey25"), guide = "none") +
  scale_x_continuous(labels = ~abs(.x), expand = expansion(add = 1),
```

```
                      name = "Number of CFMGs") +
  scale_y_discrete(expand = expansion(add = 0.5)) +
  scale_fill_manual(values = c("No" = "#E07B54", "Yes" = "#2A9D8F")) +
  coord_cartesian(clip = "off") +
  labs(y = NULL, fill = NULL,
       title = "Collaboration with private enterprise") +
  collab_theme

p1
```

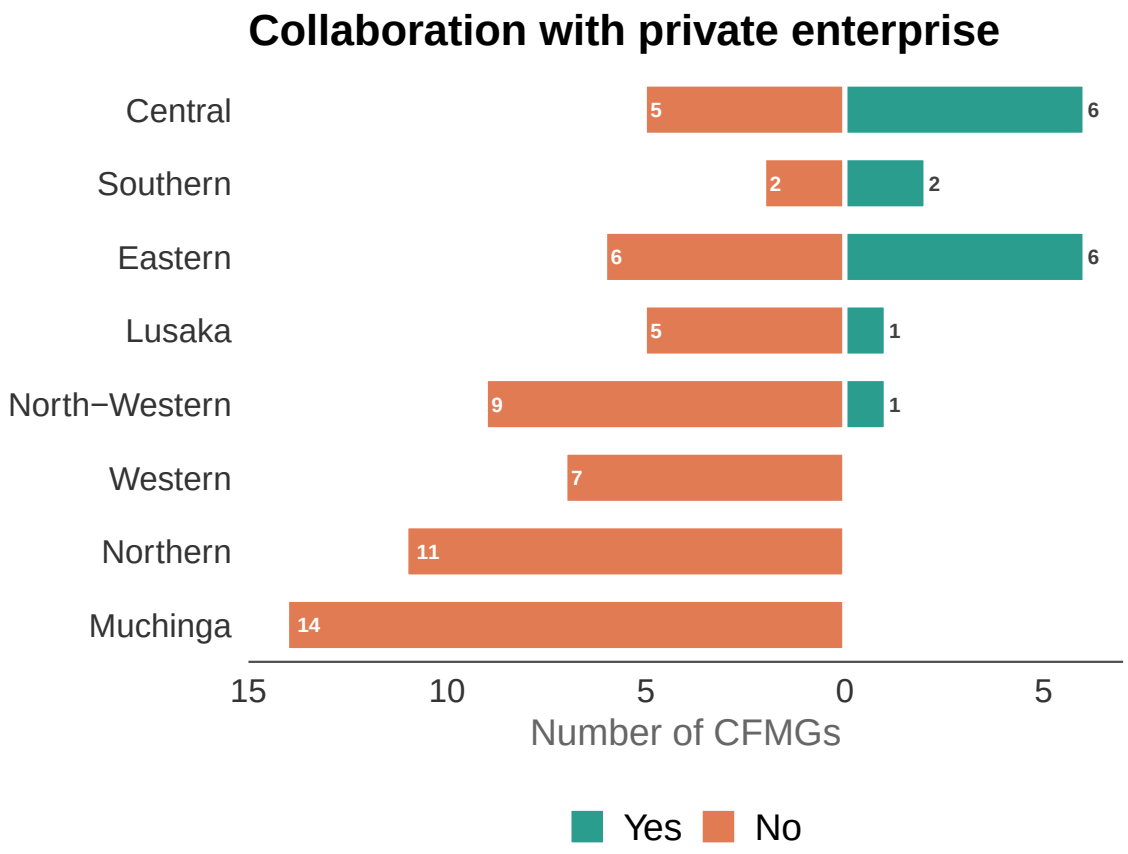

Figure 16: CFMG collaboration with private enterprise as reported by executive FGDs by province.

The value chains that CFMGs reported working with through private enterprise partnerships are shown in Figure 17. Sixteen out of 17 involved partnerships with honey off-takers and just one involved mushrooms.

```r
prefix_vc <- "economic_transformation_private_enterprises_cfmg_works_with_"

vc_labels <- c(
  "timber_companies"               = "Timber",
  "sawmills"                       = "Sawmills",
  "furniture_manufacturers"        = "Furniture",
  "wood_processing_enterprises"    = "Wood processing",
  "eco_tourism_operators"          = "Eco-tourism",
  "mushroom_processors_traders"    = "Mushrooms",
  "wild_fruit_processors_traders"  = "Wild fruit",
  "insect_processors_traders"      = "Insects",
  "chikanda_processors_traders"    = "Chikanda",
 # "carbon_credit_traders"          = "Carbon credits",
  "beekeeping_and_honey_producers" = "Honey",
  "charcoal"                       = "Charcoal",
  "botanicals_harvesting"          = "Botanicals"
)

full_vc_vars <- paste0(prefix_vc, names(vc_labels))

vc_freq <- cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(all_of(full_vc_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(label = vc_labels[sub(prefix_vc, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(label, freq)

ggplot(vc_freq, aes(x = freq, y = reorder(label, freq))) +
  geom_col(fill = "#2A9D8F", width = 0.6) +
  geom_text(aes(label = freq), hjust = -0.3, size = 4.5, colour = "grey30") +
  scale_x_continuous(
    expand = expansion(mult = c(0, 0.15)),
    name   = "Number of CFMGs"
  ) +
  scale_y_discrete(expand = expansion(add = 0.5)) +
  labs(y = NULL) +
  theme_minimal(base_size = 18) +
  theme(
    panel.grid.major.y = element_blank(),
    panel.grid.minor   = element_blank(),
```

```
    panel.grid.major.x = element_blank(),
    axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
    axis.text          = element_text(colour = "grey20"),
    plot.margin        = margin(10, 25, 10, 10)
  )
```

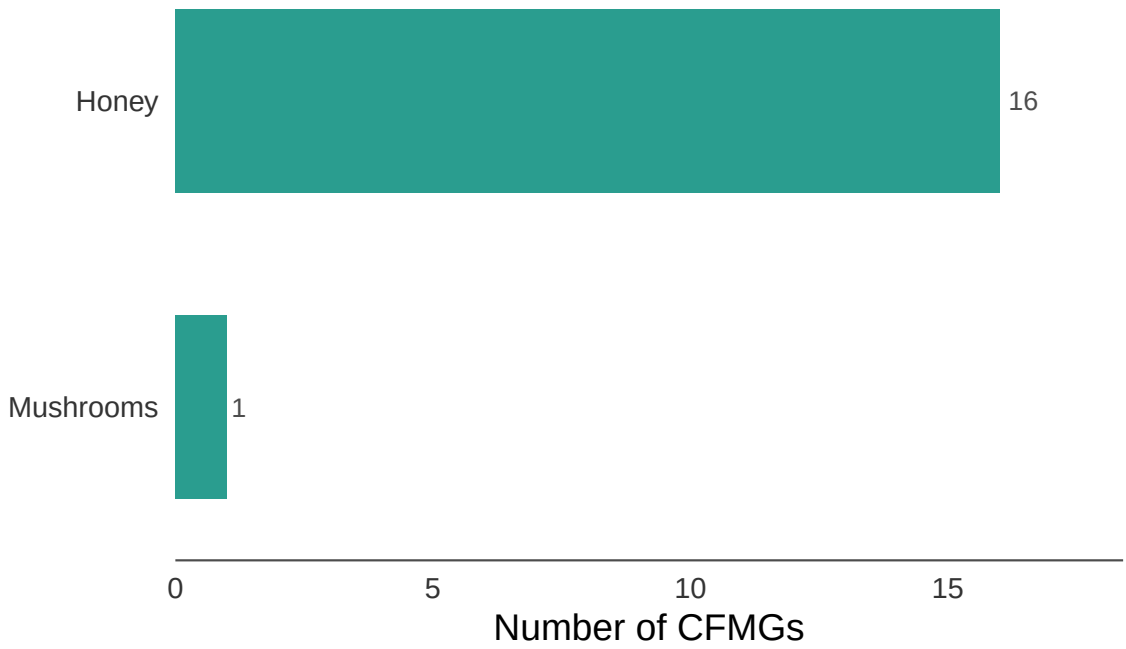

Figure 17: Value chains that CFMGs collaborate with through private enterprise partnerships, as reported by executive FGDs.

We asked the community female, male, and youth FGD groups to identify economic opportunities that remained unrealised within their CFs (Figure 18). Interestingly, although a wide range of different natural resource related value chains were mentioned, timber was mention the most often. This is noteworthy as none of the CFMGs we surveyed are involved in timber production.

```
prefix_ur <- "economic_transformation_economic_oppportunities_unrealized_economic_opportunit

ur_labels <- c(
  "timber"                    = "Timber",
  "firewood"                  = "Firewood",
```

```r
  "charcoal"                  = "Charcoal",
  "honey"                     = "Honey",
  "mushrooms"                 = "Mushrooms",
  "wild_fruits"               = "Wild fruits",
  "caterpillars"              = "Caterpillars",
  "chikanda"                  = "Chikanda",
  "munkoyo"                   = "Munkoyo",
  "lusala_busala"             = "Lusala/Busala",
  "mungongo"                  = "Mungongo",
  "water"                     = "Water",
  "devils_claw"               = "Devil's claw",
  "fish"                      = "Fish",
  "wildlife"                  = "Wildlife",
  "minerals_mining"           = "Minerals",
  "clay"                      = "Clay",
  "sand"                      = "Sand",
  "grass"                     = "Grass",
  "livestock"                 = "Livestock",
  "crops"                     = "Crops",
  "tourism_education_tours"   = "Tourism",
  "permitting"                = "Permitting",
  "infrastruture_utilities"   = "Infrastructure",
  "bamboo"                    = "Bamboo",
  "medicinal_plants"          = "Medicinal plants",
  "bushmeat"                  = "Bushmeat",
  "resins_gums"               = "Resins",
  "thatching_grass"           = "Thatching grass",
  "fibers"                    = "Fibres",
  "eco_tourism"               = "Eco-tourism"
)

full_ur_vars <- paste0(prefix_ur, names(ur_labels))

ur_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_ur_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = ur_labels[sub(prefix_ur, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)
```

```r
set.seed(42)
ggplot(ur_freq, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

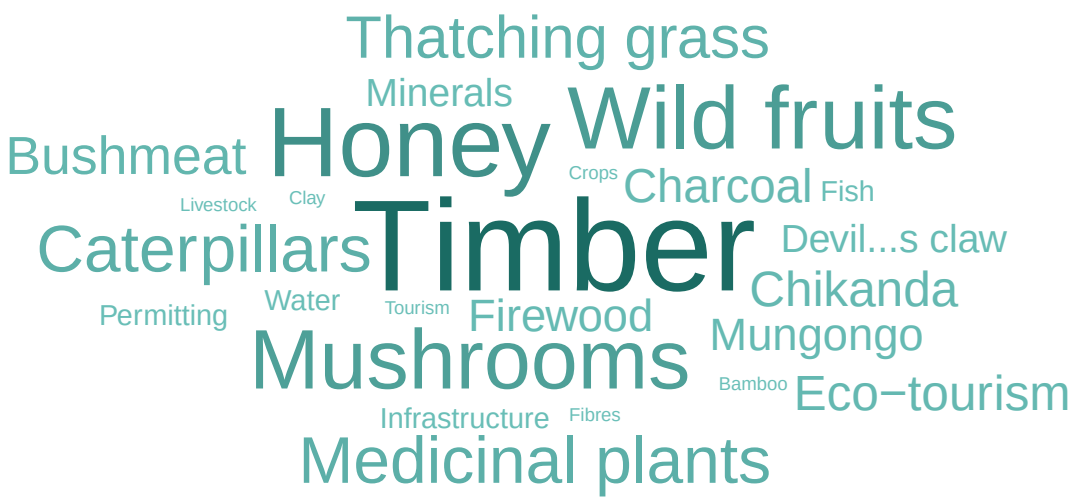

Figure 18: Unrealised economic opportunities identified by FGD groups.

The ranking of these unrealised opportunities is shown in Figure 19. Again a wide range of resources are mentioned, but the top three are timber, honey and mushrooms. Again, tt is interesting that community groups consider their timber resources to have high unrealised potential for commercial development.

```r
prefix_ur_r <- "economic_transformation_economic_oppportunities_rank_"

ur_rank_labels <- c(
  "timber"                    = "Timber",
  "firewood"                  = "Firewood",
```

```r
  "charcoal"                  = "Charcoal",
  "honey"                     = "Honey",
  "mushrooms"                 = "Mushrooms",
  "wild_fruits"               = "Wild fruits",
  "caterpillars"              = "Caterpillars",
  "chikanda"                  = "Chikanda",
  "munkoyo"                   = "Munkoyo",
  "lusala_busala"             = "Lusala/Busala",
  "mungongo"                  = "Mungongo",
  "water"                     = "Water",
  "devils_claw"               = "Devil's claw",
  "fish"                      = "Fish",
  "wildlife"                  = "Wildlife",
  "minerals_mining"           = "Minerals",
  "clay"                      = "Clay",
  "sand"                      = "Sand",
  "grass"                     = "Grass",
  "livestock"                 = "Livestock",
  "crops"                     = "Crops",
  "tourism_education_tours"   = "Tourism",
  "permitting"                = "Permitting",
  "bamboo"                    = "Bamboo",
  "medicinal_plants"          = "Medicinal plants",
  "bushmeat"                  = "Bushmeat",
  "resins_gums"               = "Resins",
  "thatching_grass"           = "Thatching grass",
  "fibers"                    = "Fibres",
  "eco_tourism"               = "Eco-tourism"
)

full_ur_rank_vars <- paste0(prefix_ur_r, names(ur_rank_labels))

cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_ur_rank_vars)) |>
  pivot_longer(everything(), names_to = "variable", values_to = "rank_val") |>
  mutate(
    rank_val = as.numeric(rank_val),
    source   = ur_rank_labels[sub(prefix_ur_r, "", variable, fixed = TRUE)]
  ) |>
  filter(rank_val %in% c(1, 2, 3)) |>
  mutate(rank = factor(paste0("Rank ", rank_val), levels = c("Rank 3", "Rank 2", "Rank 1")))
```

```r
count(source, rank) |>
group_by(source) |>
mutate(total = sum(n)) |>
ungroup() |>
ggplot(aes(y = reorder(source, total), x = n, fill = rank)) +
geom_col(width = 0.7, colour = "white", linewidth = 0.3) +
geom_text(
  aes(label = n),
  position  = position_stack(vjust = 0.5),
  size      = 3.5, colour = "white", fontface = "bold"
) +
scale_fill_manual(
  values = c("Rank 1" = "#264653", "Rank 2" = "#E9C46A", "Rank 3" = "#E07B54"),
  guide  = guide_legend(reverse = TRUE)
) +
scale_x_continuous(
  expand = expansion(mult = c(0, 0.05)),
  name   = "Number of responses"
) +
scale_y_discrete(expand = expansion(add = 0.5)) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text          = element_text(colour = "grey20"),
  legend.position    = "bottom",
  plot.margin        = margin(10, 15, 10, 10)
)
```

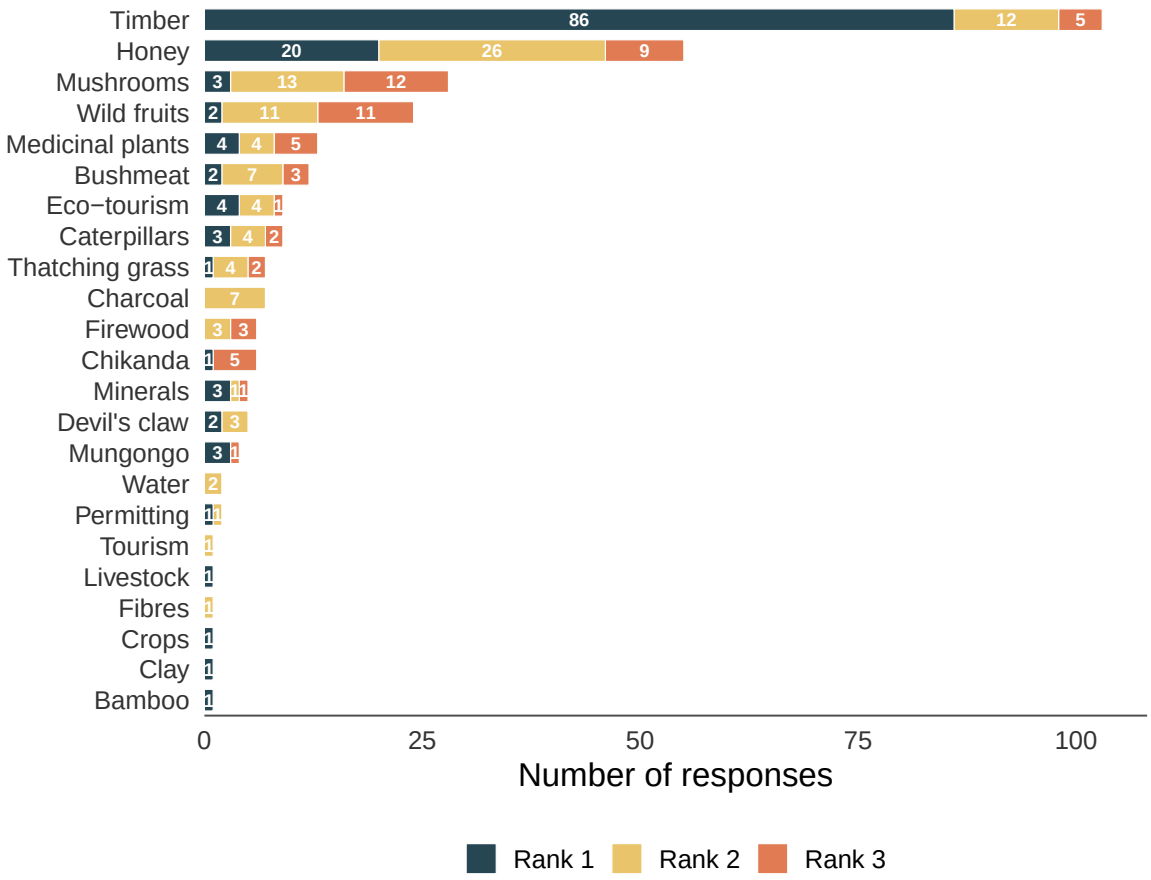

Figure 19: Ranking of unrealised economic opportunities by FGD groups. Bars show the frequency of rank 1 (highest priority), rank 2, and rank 3 responses.

FGD groups also identified potential value-added opportunities from CFM resources (Figure 20). The overlap with the previous two questions is substantial, but it is interesting that the communities specifically identify honey processing and drying mushrooms, as well as timber processing. This suggests that the communities are aware that there value addition opportunities that they are currently not realising.

```
prefix_va <- "economic_transformation_potential_value_added_opportunities_"

va_labels <- c(
  "timber_processing_e_g_furniture_carpentry_woodcrafts"
  "charcoal_briquetting_eco_friendly_alternatives_to_traditional_charcoal"
  "honey_processing_and_packaging_e_g_branded_honey_beeswax_products"
  "mushroom_drying_and_packaging_for_extended_shelf_life_and_market_access"
```

```
  "wild_fruit_processing_e_g_juices_jams_dried_fruits"
  "caterpillar_preservation_and_packaging_e_g_dried_or_canned_for_sale"
  "chikanda"
  "munkoyo"
  "lusala_busala"
  "mungongo"
  "devils_claw"
  "palm_leaves"
  "grass"
  "fish"
  "minerals"
  "bamboo_products_e_g_furniture_handicrafts_construction_materials"
  "medicinal_plant_extraction_e_g_herbal_teas_essential_oils_medicinal_creams"
  "handicrafts_and_woven_products_e_g_baskets_mats_textiles"
  "resin_and_gum_processing_e_g_adhesives_fragrances"
  "ecotourism_development_e_g_eco_lodges_cultural_experiences_tour_guiding"
  "agroforestry_based_businesses_e_g_tree_nurseries_sustainable_agriculture"
)

full_va_vars <- paste0(prefix_va, names(va_labels))

va_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_va_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = va_labels[sub(prefix_va, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

set.seed(42)
ggplot(va_freq, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

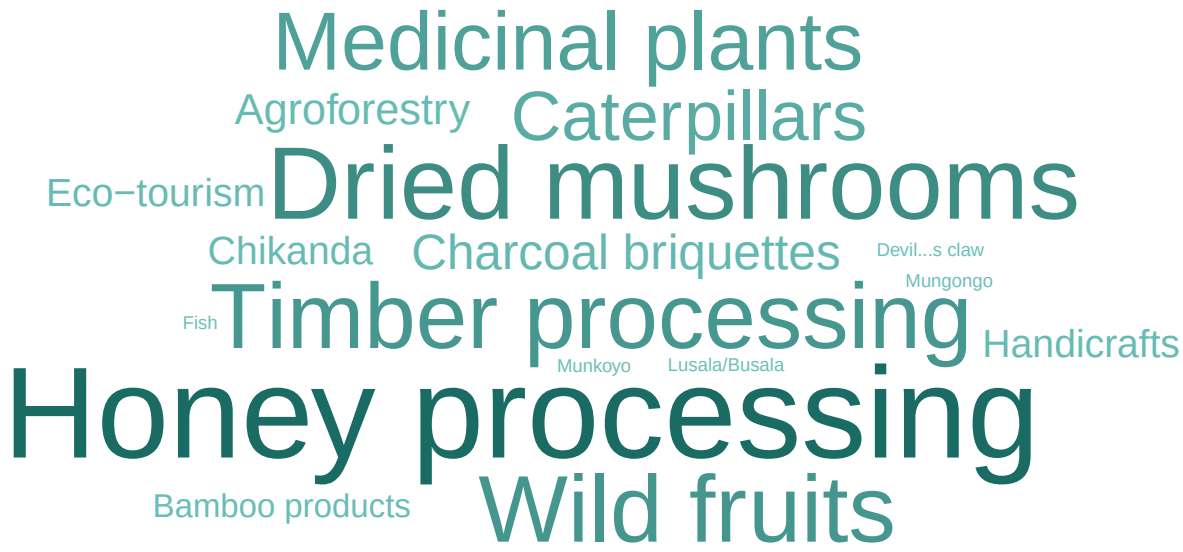

Figure 20: Potential value-added opportunities identified by FGD groups.

We asked the CFMG Executive FGDs to report which primary markets CFM products are sold into (Figure 21). Evidently, most products are sold only at the local market level, with some also being transported to national markets.

```
market_labels <- c(
  "economic_transformation_cfm_products_primary_market_local_markets"    = "Local",
  "economic_transformation_cfm_products_primary_market_national_markets" = "National",
  "economic_transformation_cfm_products_primary_market_regional_markets" = "Regional",
  "economic_transformation_cfm_products_primary_market_eu_markets"       = "EU"
)

cfmg |>
  filter(interview_type == "cfmg_executive") |>
  select(all_of(names(market_labels))) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "n") |>
  mutate(market = factor(market_labels[variable],
                         levels = c("Local", "National", "Regional", "EU"))) |>
  ggplot(aes(x = n, y = market)) +
```

```r
geom_col(fill = "#2A9D8F", width = 0.55) +
geom_text(aes(label = n), hjust = -0.3, size = 5, colour = "grey30") +
scale_x_continuous(
  expand = expansion(mult = c(0, 0.15)),
  name   = "Number of CFMGs"
) +
scale_y_discrete(limits = rev) +
labs(y = NULL) +
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text          = element_text(colour = "grey20"),
  plot.margin        = margin(10, 25, 10, 10)
)
```

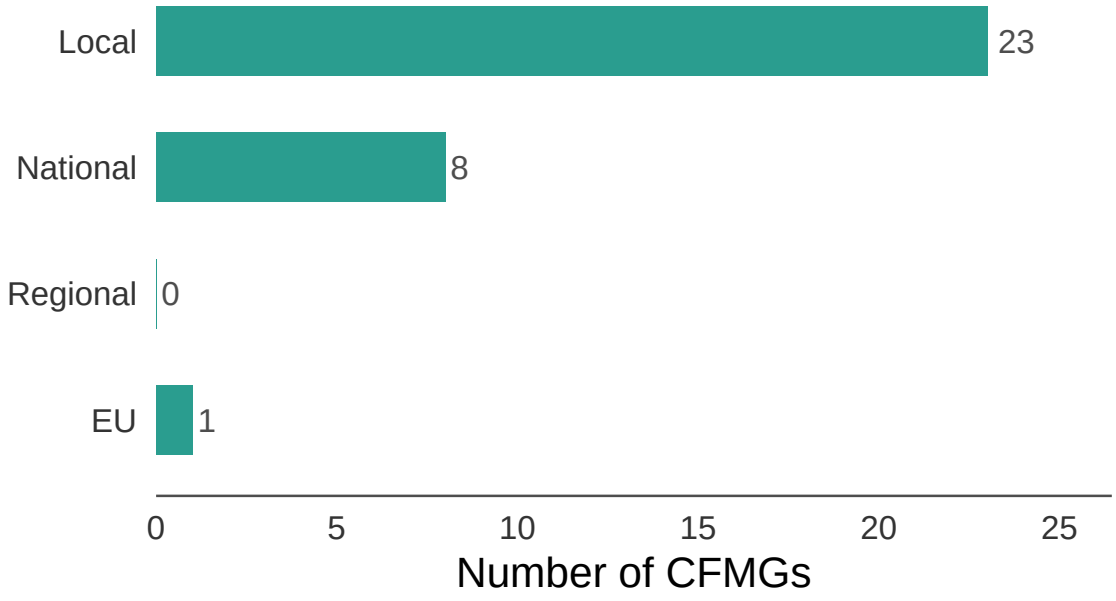

Figure 21: Primary markets for CFM products, as reported by executive FGDs.

Opportunity costs

Precluded activities

The community female, male, and youth FGD groups identified economic activities that are precluded under CFM regulations (Figure 22). While many of these are self evident, the fact that NTFP harvesting is listed quite prominently is surprising. Whether this reflects a misunderstanding on the part of the communities or indeed that they are being told they cannot collect NTFPs is not clear.

```
prefix_p <- "economic_transformation_economic_activities_precluded_cfm_"

precluded_labels <- c(
  "logging"                              = "Logging",
  "timber_harvesting"                    = "Timber harvesting",
  "hunting"                              = "Hunting",
  "charcoal_production"                  = "Charcoal",
  "commercial_mining"                    = "Mining",
  "farming"                              = "Farming",
  "livestock_grazing"                    = "Livestock grazing",
  "land_conversion"                      = "Land conversion",
  "introduction_of_invasive_species"     = "Invasive species",
  "ntfp_harvesting_of_forest_products"   = "NTFP harvesting",
  "fishing"                              = "Fishing",
  "beekeping_tradtional"                 = "Beekeeping",
  "fire_based_livelihood_activities"     = "Fire livelihoods"
)

full_p_vars <- paste0(prefix_p, names(precluded_labels))

p_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_p_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = precluded_labels[sub(prefix_p, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

set.seed(42)
ggplot(p_freq, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
```

```
  scale_colour_gradient(low = "#F4A582", high = "#8B0000") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

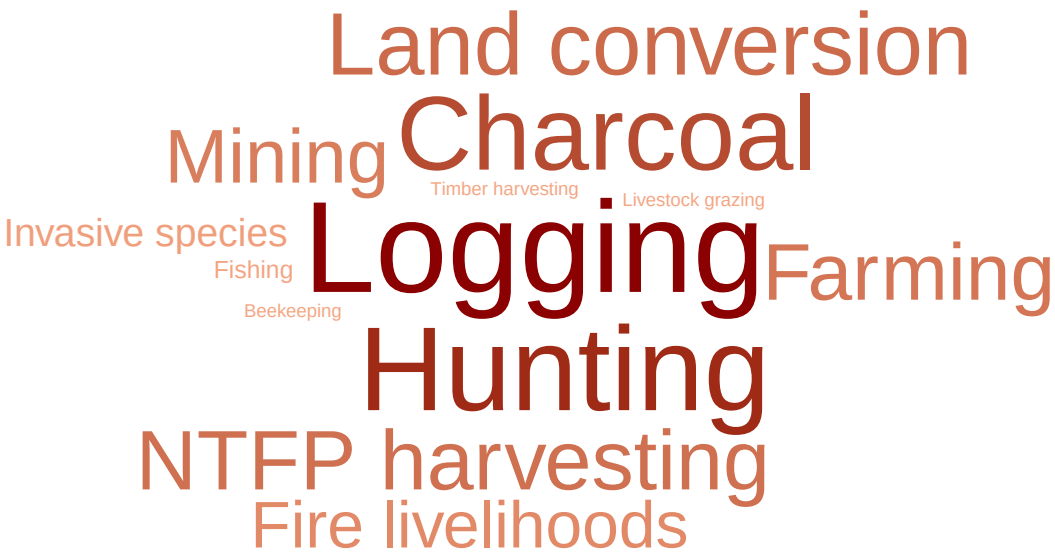

Figure 22: Economic activities precluded by CFM, as reported by FGD groups.

Other opportunity costs

FGD groups were asked whether CFM entails opportunity costs for the community (Figure 23). Interestingly, many of the FGDs did not recognise any opportunity costs. However, about a third of the FGDs did. The proportion of FGDs recognising opportunity costs was highest among community male FGDs and lowest among community female FGDs. However, none of the pairwise comparisons were significant.

```
cfmg |>
  filter(interview_type %in% c("cfmg_executive", "cfmg_female",
                               "cfmg_male", "cfmg_youth")) |>
  transmute(
    group = case_match(
      interview_type,
```

```
    "cfmg_executive" ~ "Executive",
    "cfmg_female"    ~ "Female",
    "cfmg_male"      ~ "Male",
    "cfmg_youth"     ~ "Youth"
  ),
  response = case_when(
    economic_transformation_opportunity_costs_of_cfm == "yes"          ~ "Yes",
    economic_transformation_opportunity_costs_of_cfm == "no"           ~ "No",
    str_starts(economic_transformation_opportunity_costs_of_cfm, "No_") ~ "No",
    economic_transformation_opportunity_costs_of_cfm %in% c(
      "Community forest situated very far from community",
      "The Major livelihood for this community is fishing. They don't depend so much on the
    ) ~ "No",
    TRUE ~ NA_character_
  )
) |>
drop_na(response) |>
mutate(
  group    = factor(group, levels = c("Youth", "Male", "Female", "Executive")),
  response = factor(response, levels = c("Yes", "No"))
) |>
count(group, response) |>
mutate(n_signed = if_else(response == "No", -n, n)) |>
ggplot(aes(y = group, x = n_signed, fill = response)) +
geom_col(width = 0.6, colour = "white", linewidth = 0.4) +
geom_vline(xintercept = 0, colour = "white", linewidth = 1.2) +
geom_text(
  aes(label  = n,
      hjust  = if_else(response == "No", -0.4, -0.4),
      colour = response),
  size = 4, fontface = "bold"
) +
scale_colour_manual(values = c("No" = "white", "Yes" = "grey25"),
                    guide = "none") +
scale_fill_manual(values = c("No" = "#E07B54", "Yes" = "#2A9D8F"),
                  guide  = guide_legend(reverse = TRUE)) +
scale_x_continuous(labels = ~abs(.x),
                   expand = expansion(add = 3),
                   name   = "Number of FGDs") +
scale_y_discrete(expand = expansion(add = 0.5)) +
coord_cartesian(clip = "off") +
labs(y = NULL, fill = NULL) +
```

```r
theme_minimal(base_size = 18) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text          = element_text(colour = "grey20"),
  legend.position    = "bottom",
  plot.margin        = margin(10, 25, 10, 10)
)
```

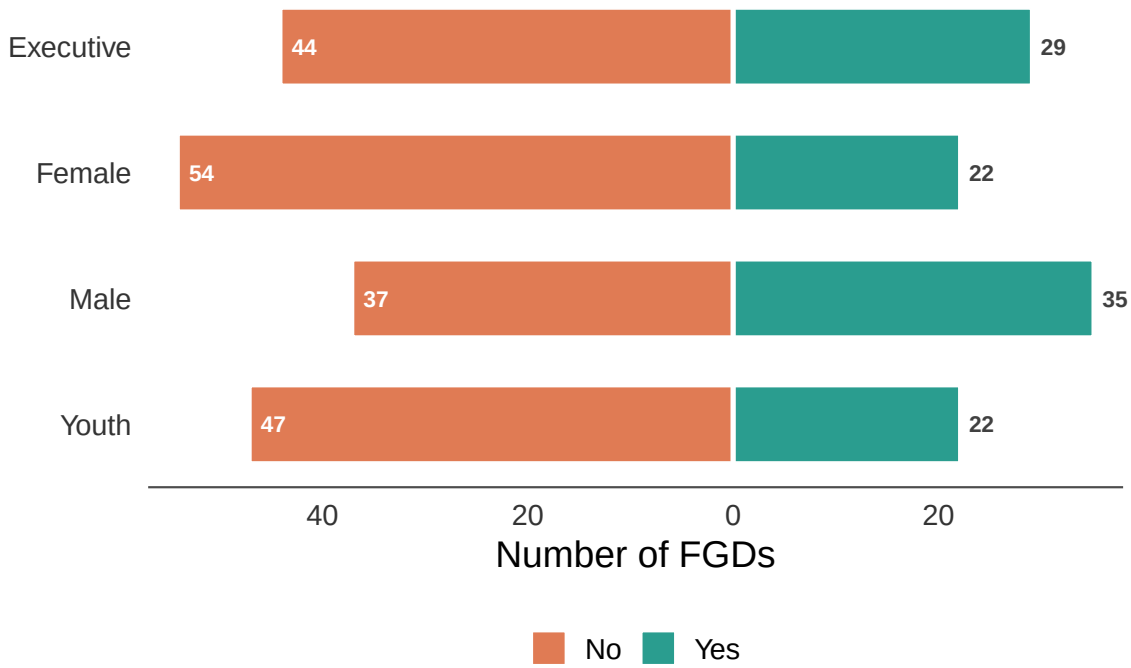

Figure 23: Whether CFM entails opportunity costs for the community, by FGD group.

Opportunity costs – FGDs – Model methods

To examine differences among FGD types in whether they reported that CFM entails opportunity costs for the community, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We used FGD type as the factor. We examined the significance of pairwise differences among FGD types using Tukey HSD.

```r
test.data.opp <- cfmg |>
  filter(interview_type %in% c("cfmg_executive", "cfmg_female",
                               "cfmg_male", "cfmg_youth")) |>
  transmute(
    province, cfmg_name,
    interview_type = factor(
      case_match(
        interview_type,
        "cfmg_executive" ~ "Executive",
        "cfmg_female"    ~ "Female",
        "cfmg_male"      ~ "Male",
        "cfmg_youth"     ~ "Youth"
      ),
      levels = c("Executive", "Female", "Male", "Youth")
    ),
    empl = case_when(
      economic_transformation_opportunity_costs_of_cfm == "yes"           ~ 1L,
      economic_transformation_opportunity_costs_of_cfm == "no"            ~ 0L,
      str_starts(economic_transformation_opportunity_costs_of_cfm, "No_") ~ 0L,
      economic_transformation_opportunity_costs_of_cfm %in% c(
        "Community forest situated very far from community",
        "The Major livelihood for this community is fishing. They don't depend so much on the
      ) ~ 0L,
      TRUE ~ NA_integer_
    )
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M9 <- glmer(empl ~ interview_type + (1 | province) + (1 | province_cfmg),
            data = test.data.opp, family = binomial)

emm.M9      <- emmeans(M9, specs = ~ interview_type, type = "response")
pairwise.M9 <- pairs(emm.M9, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interview_type + (1 | province) + (1 | province_cfmg)
   Data: test.data.opp

     AIC      BIC   logLik deviance df.resid
```

```
   371.1    393.1   -179.6    359.1      284

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.5418 -0.7302 -0.4952  0.9103  2.7612

Random effects:
 Groups        Name        Variance Std.Dev.
 province_cfmg (Intercept) 0.06352  0.252
 province      (Intercept) 0.69053  0.831
Number of obs: 290, groups:  province_cfmg, 75; province, 8

Fixed effects:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)           -0.3628     0.3942  -0.920    0.357
interview_typeFemale  -0.6043     0.3705  -1.631    0.103
interview_typeMale     0.3067     0.3588   0.855    0.393
interview_typeYouth   -0.3805     0.3726  -1.021    0.307

Correlation of Fixed Effects:
            (Intr) intr_F intr_M
intrvw_typF -0.442
intrvw_typM -0.465  0.486
intrvw_typY -0.435  0.472  0.480


 contrast           odds.ratio    SE  df null z.ratio p.value
 Executive / Female      1.830 0.678 Inf    1   1.631  0.3610
 Executive / Male        0.736 0.264 Inf    1  -0.855  0.8281
 Executive / Youth       1.463 0.545 Inf    1   1.021  0.7370
 Female / Male           0.402 0.149 Inf    1  -2.464  0.0657
 Female / Youth          0.800 0.305 Inf    1  -0.586  0.9364
 Male / Youth            1.988 0.742 Inf    1   1.841  0.2539

P value adjustment: tukey method for comparing a family of 4 estimates
Tests are performed on the log odds ratio scale
```

Individual interviewees were also asked whether CFM entails opportunity costs. Table 4 shows the percentage reporting this broken down by interviewee role. Interestingly, the community traditional leaders and community members recognise opportunity costs less of the time than other individual interviewee types, although these differences were not significant.

```r
cfmg |>
  filter(interview_type == "individual_interview") |>
  select(
    interviewee_role,
    empl = economic_transformation_opportunity_costs_of_cfm
  ) |>
  mutate(
    interviewee_role = case_match(
      as.character(interviewee_role),
      "cfmg_member"      ~ "Community member",
      "cfmg_exec_member" ~ "CFMG executive member",
      "trad_leader"      ~ "Traditional leader",
      "dfo"              ~ "District Forestry Officer",
      "NGO_rep"          ~ "NGO representative"
    ),
    interviewee_role = factor(interviewee_role, levels = c(
      "Community member", "CFMG executive member", "Traditional leader",
      "District Forestry Officer", "NGO representative"
    )),
    empl = case_when(
      empl == "yes"           ~ 1,
      empl == "no"            ~ 0,
      str_starts(empl, "No_") ~ 0,
      TRUE                    ~ NA_real_
    )
  ) |>
  group_by(interviewee_role) |>
  summarise(
    n             = n(),
    Opp_costs     = mean(empl, na.rm = TRUE),
    .groups       = "drop"
  ) |>
  arrange(interviewee_role) |>
  gt() |>
  cols_label(
    interviewee_role = "Role",
    n                = "n",
    Opp_costs        = "Report opportunity costs"
  ) |>
  fmt_percent(
    columns  = Opp_costs,
    decimals = 0
```

```
  ) |>
  tab_header(
    title    = "Opportunity costs of CFM, as reported by individual interviewees",
    subtitle = "Values represent the % of positive responses. n = the total number of interv
  ) |>
  opt_align_table_header(align = "left") |>
  cols_align(align = "left",   columns = interviewee_role) |>
  cols_align(align = "center", columns = c(n, Opp_costs)) |>
  tab_style(
    style     = cell_text(align = "center"),
    locations = cells_column_labels(columns = interviewee_role)
  ) |>
  sub_missing()
```

Table 4: Percentage of individual interviewees reporting that CFM entails opportunity costs for the community, by interviewee role.

Opportunity costs of CFM, as reported by individual interviewees

Values represent the % of positive responses. n = the total number of interview responses. A positive response indicates that the respondent considers CFM to entail opportunity costs for the community.

Role	n	Report opportunity costs
Community member	31	47%
CFMG executive member	63	51%
Traditional leader	75	35%
District Forestry Officer	56	39%
NGO representative	32	59%

Opportunity costs – Individual interviewees – Model methods

To examine differences among individual interviewee roles in whether they reported that CFM entails opportunity costs for the community, we employed a generalised linear mixed model (GLMM) with a binomial response and random intercepts for province and CF nested within province. We included interviewee role, interviewee gender and interviewee age as factors. To examine pairwise differences among interviewee roles, we employed Tukey HSD.

```
test.data.opp.ind <- cfmg |>
  filter(interview_type == "individual_interview") |>
  mutate(
```

```r
    empl = case_when(
      economic_transformation_opportunity_costs_of_cfm == "yes"           ~ 1,
      economic_transformation_opportunity_costs_of_cfm == "no"            ~ 0,
      str_starts(economic_transformation_opportunity_costs_of_cfm, "No_") ~ 0,
      TRUE                                                                ~ NA_real_
    ),
    interviewee_gender = demo_information_interviewee_gender,
    interviewee_age    = demo_information_interviewee_age
  ) |>
  drop_na(empl) |>
  mutate(province_cfmg = interaction(province, cfmg_name))

M10 <- glmer(empl ~ interviewee_role + interviewee_gender + interviewee_age +
               (1 | province) + (1 | province_cfmg),
             data = test.data.opp.ind, family = binomial)

emm.M10      <- emmeans(M10, specs = ~ interviewee_role, type = "response")
pairwise.M10 <- pairs(emm.M10, adjust = "Tukey")
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
  Approximation) [glmerMod]
 Family: binomial  ( logit )
Formula: empl ~ interviewee_role + interviewee_gender + interviewee_age +
    (1 | province) + (1 | province_cfmg)
   Data: test.data.opp.ind

     AIC      BIC   logLik deviance df.resid
   339.2    370.7   -160.6    321.2      236

Scaled residuals:
    Min      1Q  Median      3Q     Max
-1.5084 -0.8786 -0.5305  0.9503  2.3170

Random effects:
 Groups        Name        Variance  Std.Dev.
 province_cfmg (Intercept) 6.055e-09 7.782e-05
 province      (Intercept) 3.848e-01 6.204e-01
Number of obs: 245, groups:  province_cfmg, 73; province, 8

Fixed effects:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                      -0.954868   0.809547  -1.180    0.238
```

```
interviewee_rolecfmg_exec_member -0.109039   0.481550  -0.226    0.821
interviewee_roletrad_leader      -0.343588   0.476762  -0.721    0.471
interviewee_roledfo              -0.202862   0.506821  -0.400    0.689
interviewee_roleNGO_rep           0.720755   0.575682   1.252    0.211
interviewee_gendermale            0.405050   0.359538   1.127    0.260
interviewee_age                   0.007464   0.012223   0.611    0.541

Correlation of Fixed Effects:
            (Intr) int___ intr__ intrvw_r i_NGO_ intrvw_gn
intrvw_rl__ -0.373
intrvw_rlt_ -0.239  0.605
intrvw_rldf -0.526  0.614  0.589
intrvw_NGO_ -0.502  0.512  0.509  0.564
intrvw_gndr -0.319 -0.020 -0.099 -0.053   -0.002
interview_g -0.760  0.012 -0.157  0.222    0.228 -0.023
optimizer (Nelder_Mead) convergence code: 0 (OK)
boundary (singular) fit: see help('isSingular')


 contrast                        odds.ratio    SE  df null z.ratio p.value
 cfmg_member / cfmg_exec_member      1.115 0.537 Inf    1   0.226  0.9994
 cfmg_member / trad_leader           1.410 0.672 Inf    1   0.721  0.9518
 cfmg_member / dfo                   1.225 0.621 Inf    1   0.400  0.9946
 cfmg_member / NGO_rep               0.486 0.280 Inf    1  -1.252  0.7206
 cfmg_exec_member / trad_leader      1.264 0.538 Inf    1   0.551  0.9818
 cfmg_exec_member / dfo              1.098 0.478 Inf    1   0.216  0.9995
 cfmg_exec_member / NGO_rep          0.436 0.230 Inf    1  -1.570  0.5166
 trad_leader / dfo                   0.869 0.388 Inf    1  -0.315  0.9979
 trad_leader / NGO_rep               0.345 0.182 Inf    1  -2.014  0.2589
 dfo / NGO_rep                       0.397 0.202 Inf    1  -1.814  0.3650

Results are averaged over the levels of: interviewee_gender
P value adjustment: tukey method for comparing a family of 5 estimates
Tests are performed on the log odds ratio scale
```

The types of opportunity costs identified are shown in Figure 24. Mentions of reduced access, land restrictions and land use conflict are by some margin the most frequently identified opportunity costs. However, it is also interesting that time costs and diverting community resources are also quite prevalent. CFM requires that community members meet which entails a time cost and community resources are diverted to managing the forest that might otherwise be spent on other activities, such as agriculture.

```r
prefix_oc <- "economic_transformation_main_opportunity_costs_experienced_"

oc_labels <- c(
  "reduced_access_to_forest_resources_e_g_firewood_building_materials"    = "Reduced access"
  "time_spent_on_cfm_activities_instead_of_income_generating_work"        = "Time cost",
  "conflict_with_other_land_resource_uses"                                = "Land use confli
  "restrictions_on_land_use_for_farming_or_other_activities"              = "Land restrictio
  "economic_costs_from_diverting_community_resources_to_cfm"              = "Diverting resou
)

full_oc_vars <- paste0(prefix_oc, names(oc_labels))

binary_oc <- cfmg |>
  select(all_of(full_oc_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = oc_labels[sub(prefix_oc, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

comment_oc <- cfmg |>
  select(text = economic_transformation_main_opportunity_costs_experienced_comment) |>
  filter(!is.na(text), trimws(text) != "") |>
  unnest_tokens(word, text) |>
  anti_join(stop_words, by = "word") |>
  mutate(word = case_match(
    word,
    c("charcoal", "production", "prohibition", "stopped",
      "restriction", "restrictions") ~ "Charcoal",
    "chitemene"                        ~ "Chitemene",
    "grazing"                          ~ "Grazing",
    "timber"                           ~ "Timber",
    "logging"                          ~ "Logging",
    "fruits"                           ~ "Fruits",
    "medicine"                         ~ "Medicinal plants",
    .default                           = word
  )) |>
  filter(!word %in% c("reduced", "involved", "traditional",
                      "interference", "access", "source",
                      "villages", "moment")) |>
  count(word, name = "freq")
```

```r
wc_oc <- bind_rows(binary_oc, comment_oc) |>
  group_by(word) |>
  summarise(freq = sum(freq), .groups = "drop")

set.seed(42)
ggplot(wc_oc, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 18)) +
  scale_colour_gradient(low = "#F4A582", high = "#8B0000") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

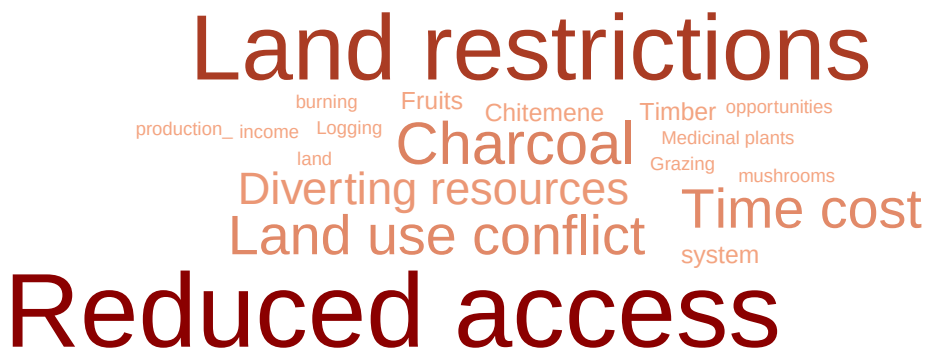

Figure 24: Types of opportunity costs experienced under CFM.

Discussion

Through the FGDs and the individual interviews, we investigated a diverse set of indicators of economic transformation. We covered access to natural resources for household use and sale, employment, income earning opportunities, including carbon finance and other sources of income, and opportunity costs.

Most FGD groups and individual interviewees indicated that CFM had not increased access to natural resources for either household use or sale. However, about a third of respondents considered that access to resources had improved under CFM. Given that the communities were involved in harvesting forest products before the instigation of CFM, it is not surprising that most respondents do not recognise any change. However, around one third of respondent groups and individuals felt that access had improved under CFM. This presumably reflects the formalisation of community rights to access forest resources.

A larger proportion of FGD groups considered that CFM had had a positive effect on full-time employment. However, the differences among different FGD types were highly significant. More community female groups suggested there was no change in employment and more community youth groups felt full-time formal employment opportunities had increased. We also showed that the numbers employed included many more men than women and, given that by far the most important employment roles were HFOs and Community Scouts, it is likely they also included a disproportionate number of youth, which would explain these differences in perception. Moreover, while CFM does appear to have increased formal employment, the numbers are quite paltry. However, this may reflect the fact that employment in forest product harvesting and processing is rarely formal or full-time. In short, while CMF does appear to have resulted in some new employment opportunities, these have been largely limited to HFO and Community Scout roles and have not benefited many women.

A large proportion of CFs in Zambia were formed on the promise of carbon dollars. However, only approximately one third of the CFs in our sample appeared to have benefited from carbon income at the time of the survey. The fact that a number of the FDG groups, especially youth groups, were unaware of the carbon agreement is concerning. It suggests that the carbon trading companies may have failed to conduct a proper FPIC process involving the entire community and that the CFMG Executives may not be communicating properly with their communities. Nevertheless, many carbon traders are clearly supporting communities. Most of the roles concern forest protection and agricultural extension for sustainable intensification, for example, climate smart agriculture. Both of these may be considered as essential to a carbon business plan. NTFPs and enterprise development were less often mentioned as a part of the carbon trader activities. This is perhaps an oversight on their part, because enhancing forest product value chains would increase the value of forest conservation for local communities.

The above results are backed up by the response to questions concerning other income earning opportunities. It seems CFM has not led to increased income from market linkages, alternative livelihoods, conservation revenue, forest product processing or ecotourism except for in a very few cases. Nevertheless, a wide variety of forest products clearly contribute to household use and some additional income. Although honey and mushrooms were most often identified as the most important source of additional income, another 10 NTFPs were mention as the most important at least once. However, these additional sources of income were identified in less than half of the CFs and only 13 identified a second rank. Thus, forest products are rarely being exploited as a source of additional income and if they are it is usually only one. Only a small number of CFMGs are partnering with an enterprise, such as an off-taker, and among

those that do it almost exclusively concerns honey. Nonetheless, the communities identified a wide diversity of forest products that could potentially be harvested and processed to provide an additional source of income. Prominent among these were timber and wild fruit, as well as mushrooms and honey. In some cases timber harvesting may conflict with a carbon business plans, but not if plantations were established. The potential of CFM to support forest product enterprises is clearly massively under appreciated.

Unfortunately, the information on incomes from CFM were not very informative. Some clearly included income from carbon while most did not and it seems that the CFMG Executives have little understanding of the value of forest products being harvested from their CFs. To get a more accurate picture of these values a more in depth PEN type survey is recommended.

The local communities identified a number of opportunity costs to CFM. Most of these pertained to activities that are precluded, such as those involving forest clearance. However, many groups also identified other opportunity costs, such as time costs and diverted resources. Recognition of opportunity costs is important to understand the cost-benefits of CFM for communities.

Overall, a positive picture of economic transformation emerges from the responses of the FGDs and individual interviews. While changes in indicators, such as employment or additional income from forest products, many not appear substantial, they are in a positive direction. Most CFMGs will also receive carbon income, even if this has only benefited around one third of CFMGs so far. However, what is also evident is that CFMG business plans are over-reliant on just carbon or sometimes carbon and honey. Substantial opportunities for developing much more diversfied business plans exist. These could include a range of different NTFPs and potentially also timber. In addition to simply increasing income earning opportunities, diversified business plans would also provide a buffer against fluctuating commodity prices and because different products are harvested at different times of the year, they can help stabilise income throughout the year as well.